

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```

nan_ratio = df[cols_to_check].isna().mean()
valid_features = nan_ratio[nan_ratio <= 0.15].index
valid_features_list = list(valid_features)

#df = df[valid_features]
# Removing of last np.nan
mask = np.isfinite(df[valid_features]).all(axis=1)
df = df[mask]

cols = list(dict.fromkeys(valid_features_list + #target_col + features_X +
                        feats_to_balance + feats_grouping +
                        feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

new_feats_X = [x for x in features_X if x in valid_features_list]
new_feats_add = [x for x in feats_eng if x in valid_features_list]

data["df"]=df
data["features_X"] = new_feats_X
data["feats_added"] = new_feats_add

column_removed = [x for x in cols_to_check if x not in valid_features_list]
print(f'Column removed because np.isnan > 15%: {column_removed}')

return data

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":

```

```

df["campaign"] = df["datetime"].apply(assign_campaign)
df = df[df["campaign"] != "other"].copy()
df["innova_point"] = df.apply(assign_innova_point, axis=1)
df = df[df["innova_point"] != "no"].copy()
with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
    #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["columnn"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}

```

```

        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")

```

Column removed because np.isnan > 15%: []

1.2 Metadata

Nombre de lignes df	7224
Nombre de lignes df clean	7224
Nombre de features	32
% de NaN	0.0
Taille dataset entraînement	5056
Taille dataset test	2167
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_24h_D1_energy, r_NH3_wl_24h_D2_energy, r_NH3_wl_24h_D3_energy, r_NH3_wl_24h_A_energy, r_NH3_wl_24h_A_rms, r_NH3_wl_24h_D1_energy_rel, r_NH3_wl_24h_D2_energy_rel, r_NH3_wl_24h_D3_energy_rel, r_NH3_wl_24h_A_energy_rel, r_temperature_wl_24h_D1_energy, r_temperature_wl_24h_D2_energy, r_temperature_wl_24h_D3_energy, r_temperature_wl_24h_A_energy, r_temperature_wl_24h_A_rms, r_temperature_wl_24h_D1_energy_rel, r_temperature_wl_24h_D2_energy_rel, r_temperature_wl_24h_D3_energy_rel, r_temperature_wl_24h_A_energy_rel, r_umidity_wl_24h_D1_energy, r_umidity_wl_24h_D2_energy, r_umidity_wl_24h_D3_energy, r_umidity_wl_24h_A_energy, r_umidity_wl_24h_A_rms, r_umidity_wl_24h_D1_energy_rel, r_umidity_wl_24h_D2_energy_rel, r_umidity_wl_24h_D3_energy_rel, r_umidity_wl_24h_A_energy_rel

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

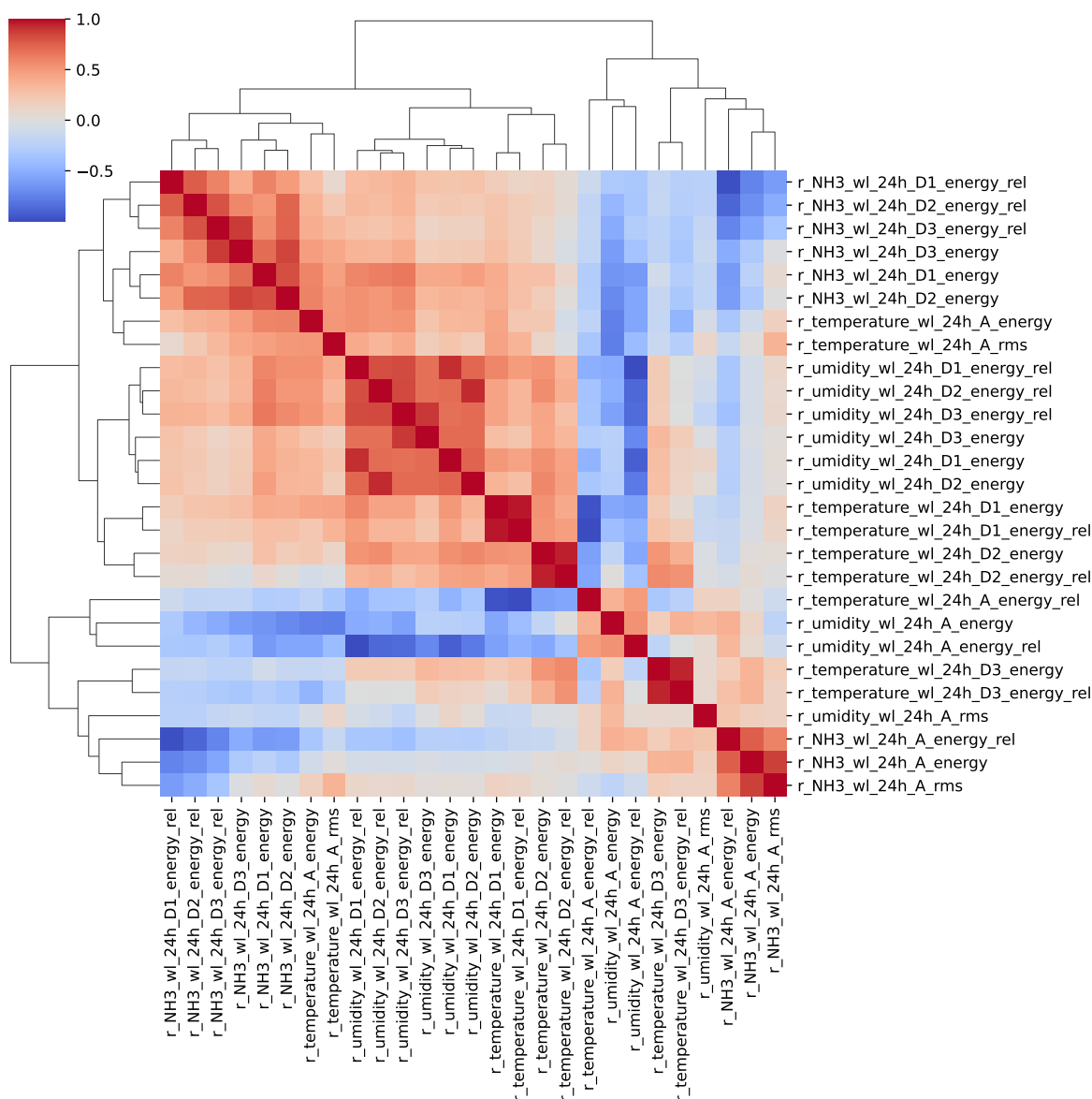
The following section has been done with a threshold choosen of : 0.8

cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, _, T, H, I ; **cluster 6:** r_NH3_wl_24h_D2_energy_rel, r_NH3_wl_24h_D3_energy_rel, r_NH3_wl_24h_D1_energy_rel, r_NH3_wl_24h_D1_energy, r_NH3_wl_24h_D3_energy, r_NH3_wl_24h_A_energy_rel, r_NH3_wl_24h_D2_energy ; **cluster 7:** r_NH3_wl_24h_A_energy, r_NH3_wl_24h_A_rms ; **cluster 8:** r_temperature_wl_24h_D1_energy, r_temperature_wl_24h_A_energy_rel, r_temperature_wl_24h_D1_energy_rel ; **cluster 9:** r_temperature_wl_24h_D2_energy_rel, r_temperature_wl_24h_D2_energy ; **cluster 10:** r_temperature_wl_24h_D3_energy_rel, r_temperature_wl_24h_D3_energy ; **cluster 11:** r_temperature_wl_24h_A_energy ; **cluster 12:** r_temperature_wl_24h_A_rms ; **cluster 13:** r_umidity_wl_24h_D3_energy, r_umidity_wl_24h_D2_energy, r_umidity_wl_24h_A_energy_rel, r_umidity_wl_24h_D1_energy, r_umidity_wl_24h_D3_energy_rel, r_umidity_wl_24h_D1_energy_rel, r_umidity_wl_24h_D2_energy_rel ; **cluster 14:** r_umidity_wl_24h_A_energy ; **cluster 15:** r_umidity_wl_24h_A_rms ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_24h_D2_energy_rel, r_NH3_wl_24h_A_energy, r_temperature_wl_24h_D1_energy, r_temperature_wl_24h_D2_energy_rel, r_temperature_wl_24h_D3_energy_rel, r_temperature_wl_24h_A_energy, r_temperature_wl_24h_A_rms, r_umidity_wl_24h_D3_energy, r_umidity_wl_24h_A_energy, r_umidity_wl_24h_A_rms



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_24h_D2_energy_rel, r_NH3_wl_24h_A_energy, r_temperature_wl_24h_D1_energy, r_temperature_wl_24h_D2_energy_rel, r_temperature_wl_24h_D3_energy_rel, r_tempera-

ture_wl_24h_A_energy, r_temperature_wl_24h_A_rms, r_umidity_wl_24h_D3_energy,
r_umidity_wl_24h_A_energy, r_umidity_wl_24h_A_rms

Len(Features_X): 15

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’:
None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, ran-
dom_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

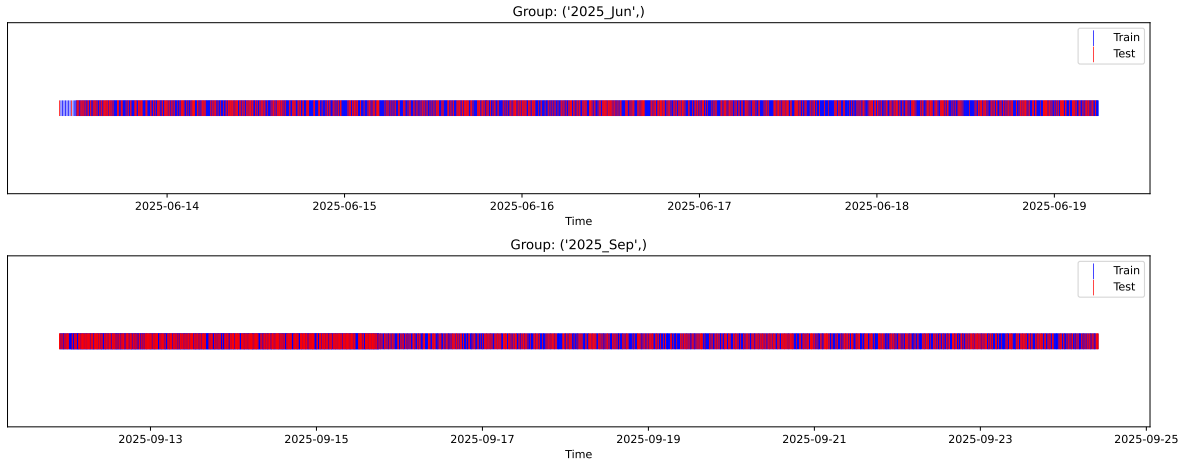
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

External Split: Train: 5056 samples Test: 2168 samples



3.3.2 Internal CV Folds

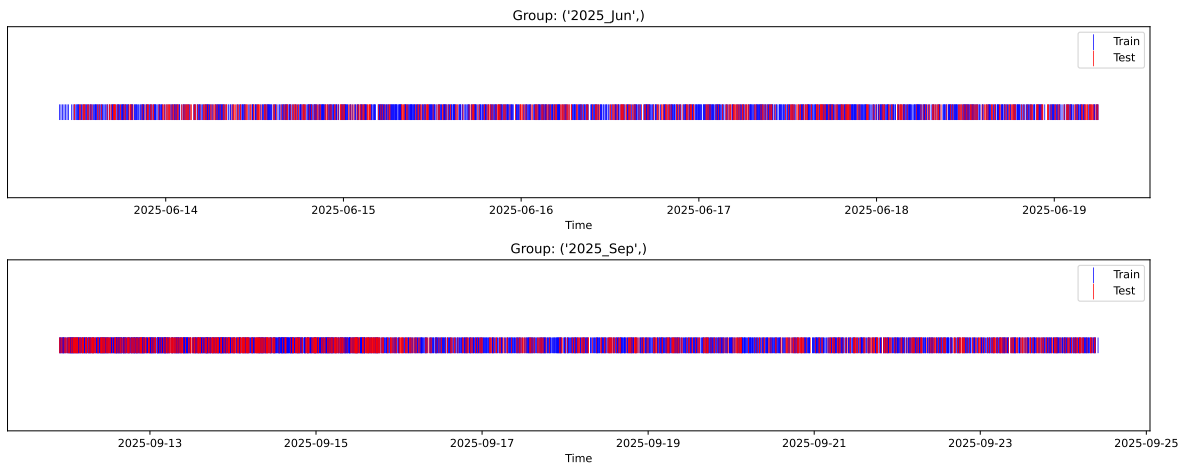
Fold 1/5

Train: 3539 samples Test: 1517 samples Excluded: 0 samples Window exclusion: 0h



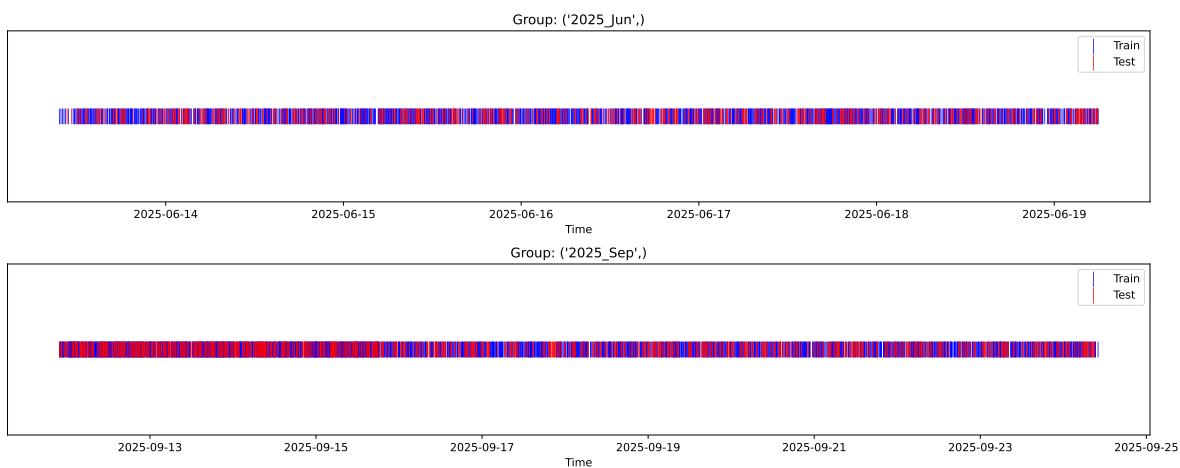
Fold 2/5

Train: 3539 samples Test: 1517 samples Excluded: 0 samples Window exclusion: 0h



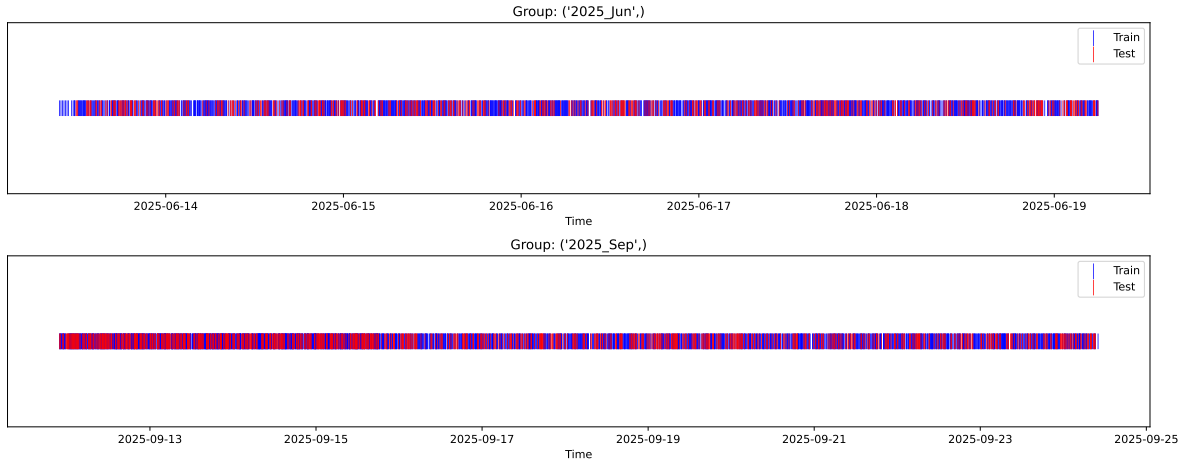
Fold 3/5

Train: 3539 samples Test: 1517 samples Excluded: 0 samples Window exclusion: 0h



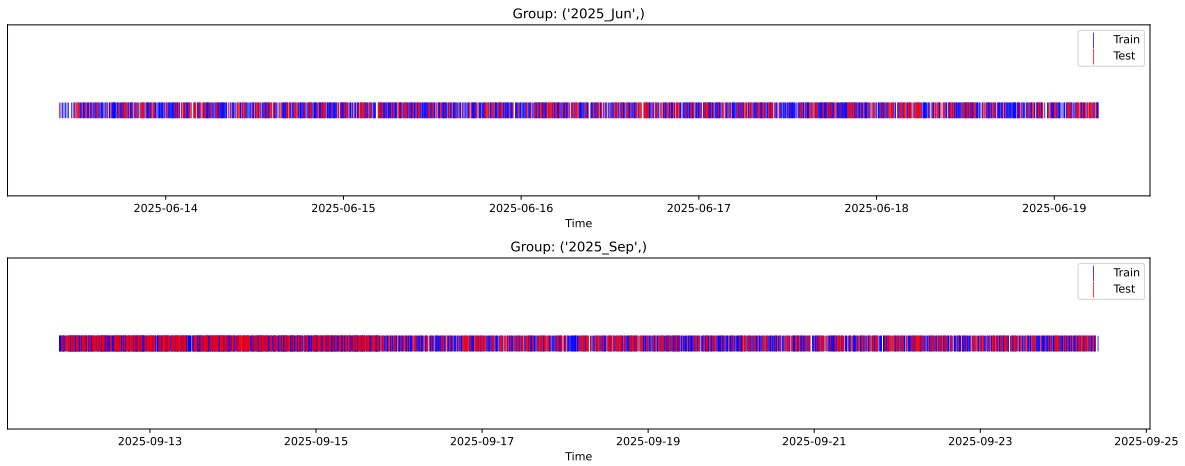
Fold 4/5

Train: 3539 samples Test: 1517 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3539 samples Test: 1517 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.346, std=0.787, min=0.959, max=8.190
 y_{test} : mean=2.382, std=0.828, min=1.071, max=13.845

Fold 0 — y_{test} : mean=2.35, std=0.80, y_{train} : mean=2.34, std=0.78

Fold 1 — y_{test} : mean=2.33, std=0.75, y_{train} : mean=2.35, std=0.80

Fold 2 — y_{test} : mean=2.37, std=0.83, y_{train} : mean=2.34, std=0.77

Fold 3 — y_test: mean=2.34, std=0.78, y_train: mean=2.35, std=0.79

Fold 4 — y_test: mean=2.32, std=0.80, y_train: mean=2.36, std=0.78

train scores CV: [0.42734793 0.41832809 0.42044213 0.42589541 0.42038063]

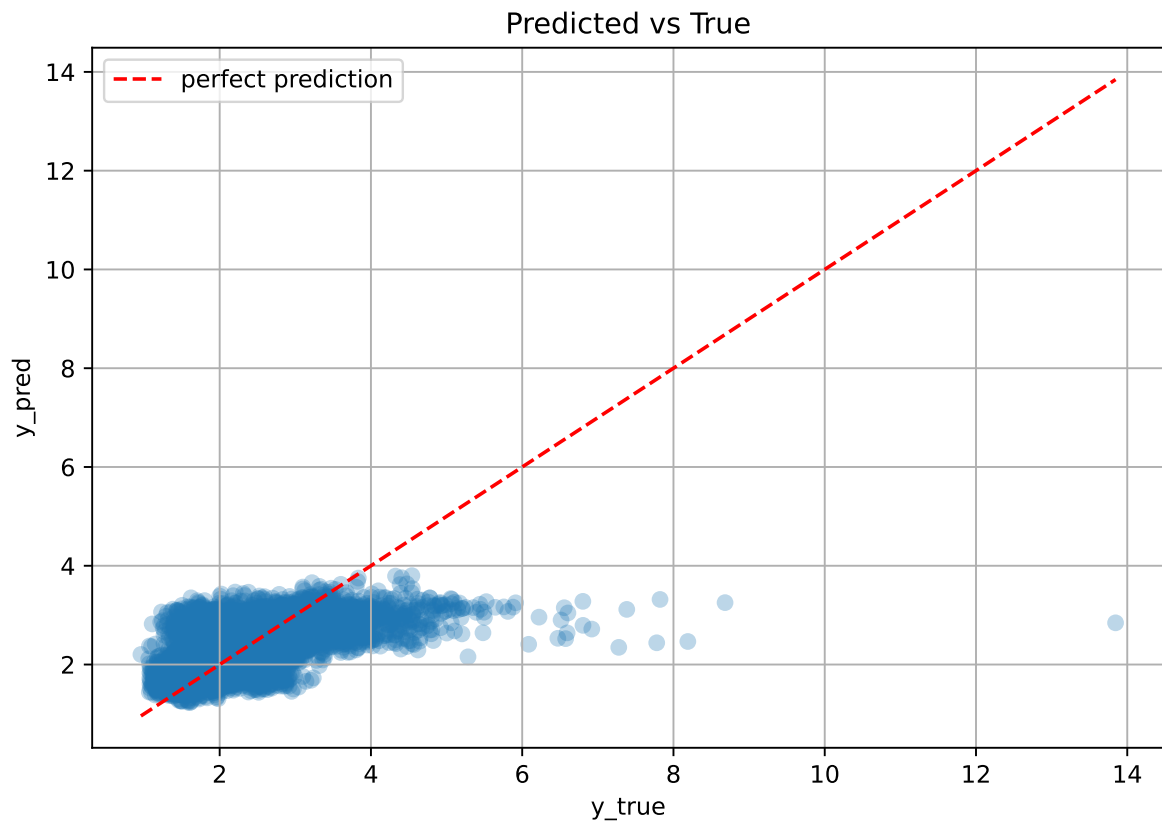
test scores CV: [0.4027994 0.42279458 0.41650604 0.40460566 0.41746292]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4225	0.4128	0.0078	0.3866	0.3887	-0.0242	0.9415	1.0234

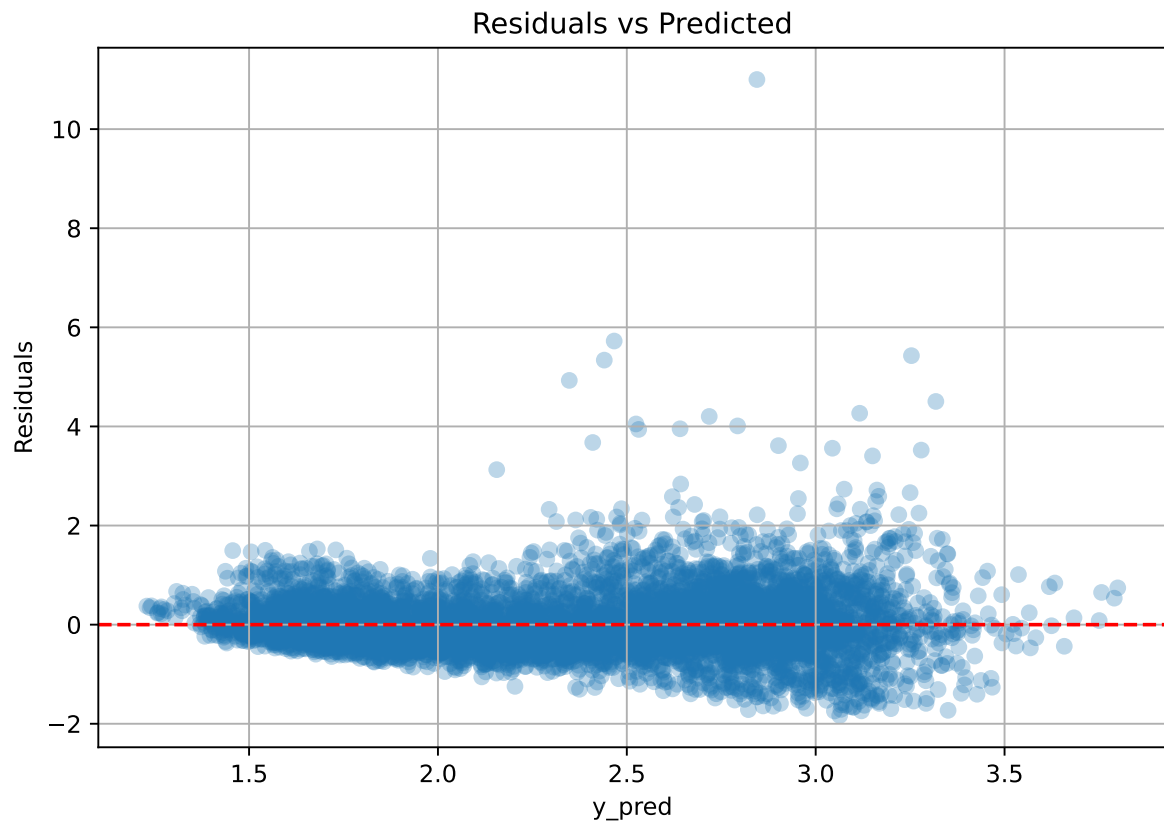
Coefficients:

coef_r_CO2 = -0.0481, coef_r_NH3 = -0.0773, coef_r_NH3_wl_24h_A_energy = -0.0324, coef_r_NH3_wl_24h_D2_energy_rel = -0.0105, coef_r_THI = 0.7323, coef_r_temperature = -0.9548, coef_r_temperature_wl_24h_A_energy = 0.0717, coef_r_temperature_wl_24h_A_rms = 0.1227, coef_r_temperature_wl_24h_D1_energy = 0.0663, coef_r_temperature_wl_24h_D2_energy_rel = -0.0012, coef_r_temperature_wl_24h_D3_energy_rel = 0.1176, coef_r_umidity = -0.7775, coef_r_umidity_wl_24h_A_energy = 0.1914, coef_r_umidity_wl_24h_A_rms = -0.0719, coef_r_umidity_wl_24h_D3_energy = 0.0488, intercept = 2.3462,

3.4.2 Scatter



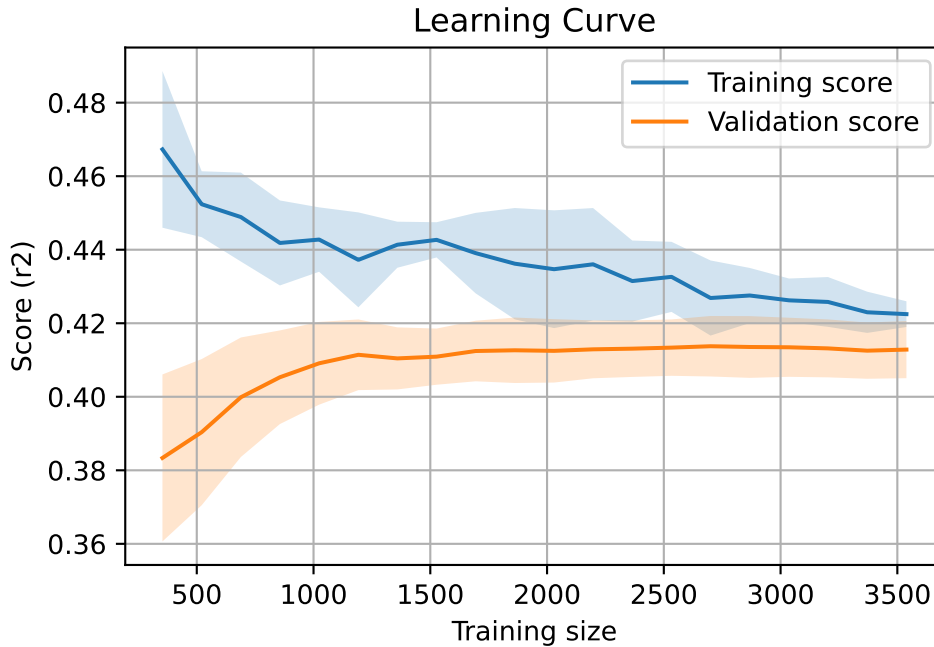
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7224, `Len(training)` : 5057,

`Len(training_real)` : 3540, `Len(test_of_training)` : 1517, `Len(test)` : 2167



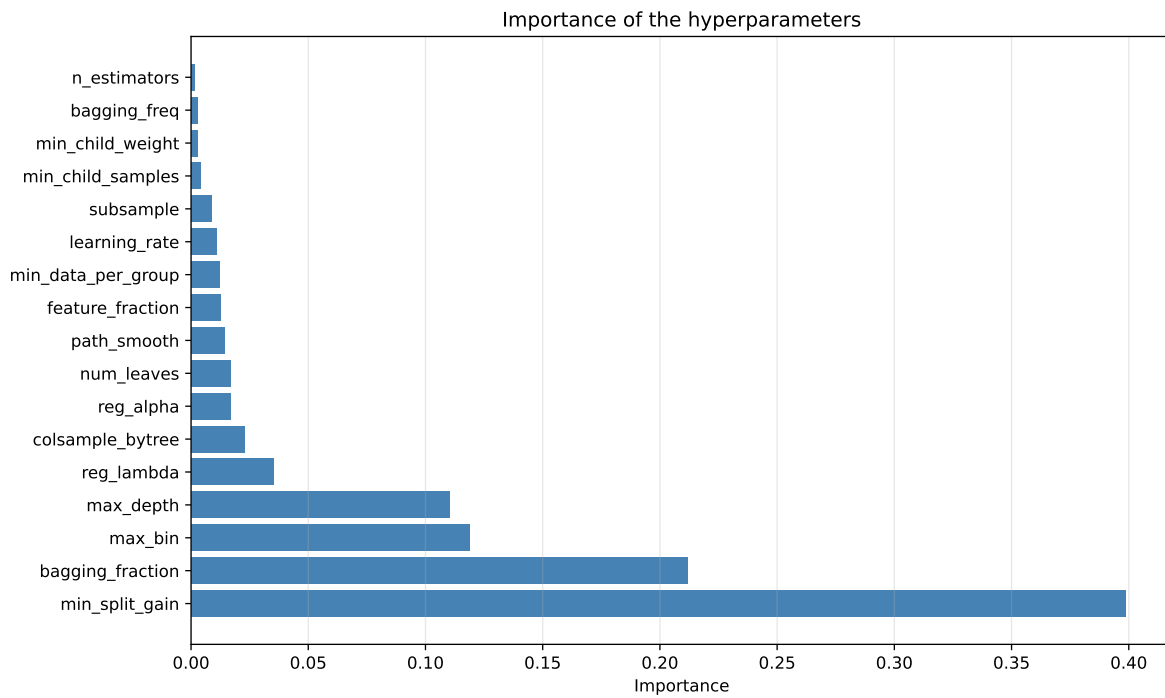
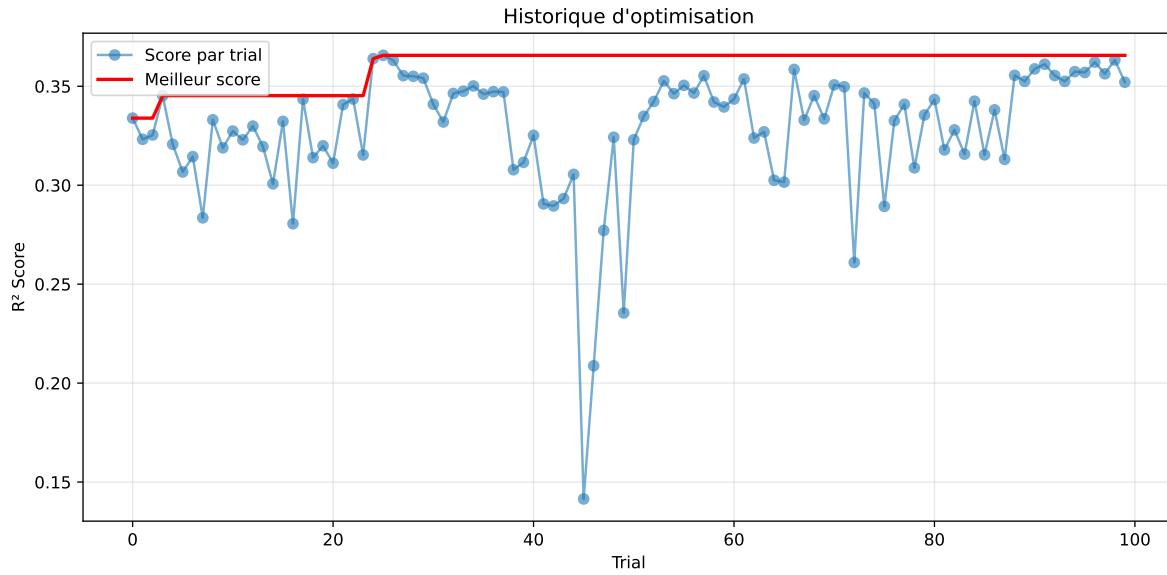
3.5 Model : LGBM

3.5.1 Gridsearch

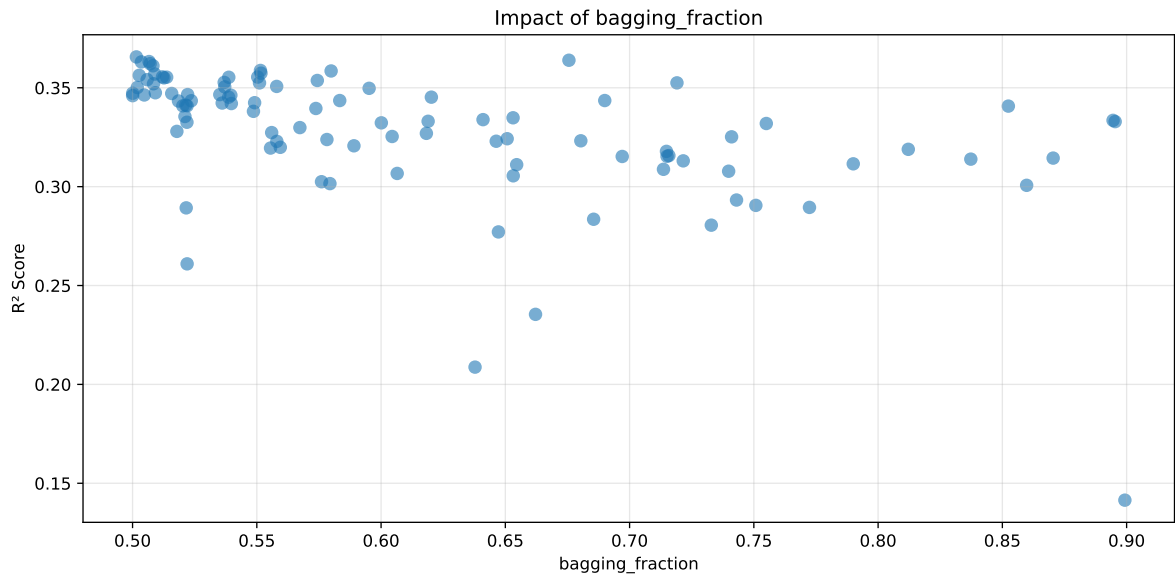
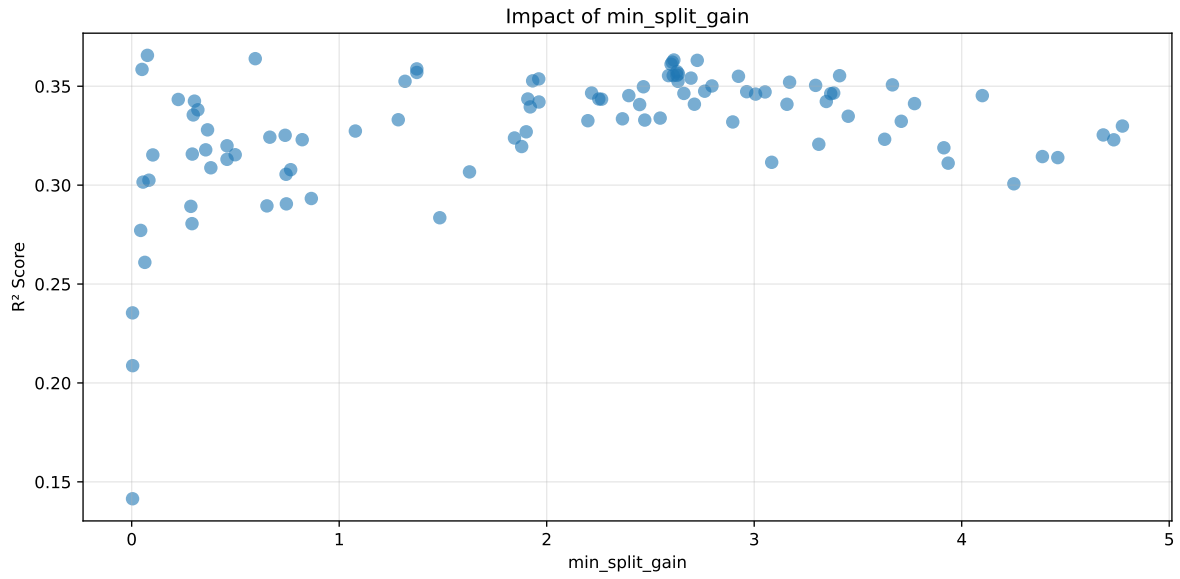
Distribution y_train vs y_test: y_train: mean=2.346, std=0.787, min=0.959, max=8.190
y_test: mean=2.382, std=0.828, min=1.071, max=13.845

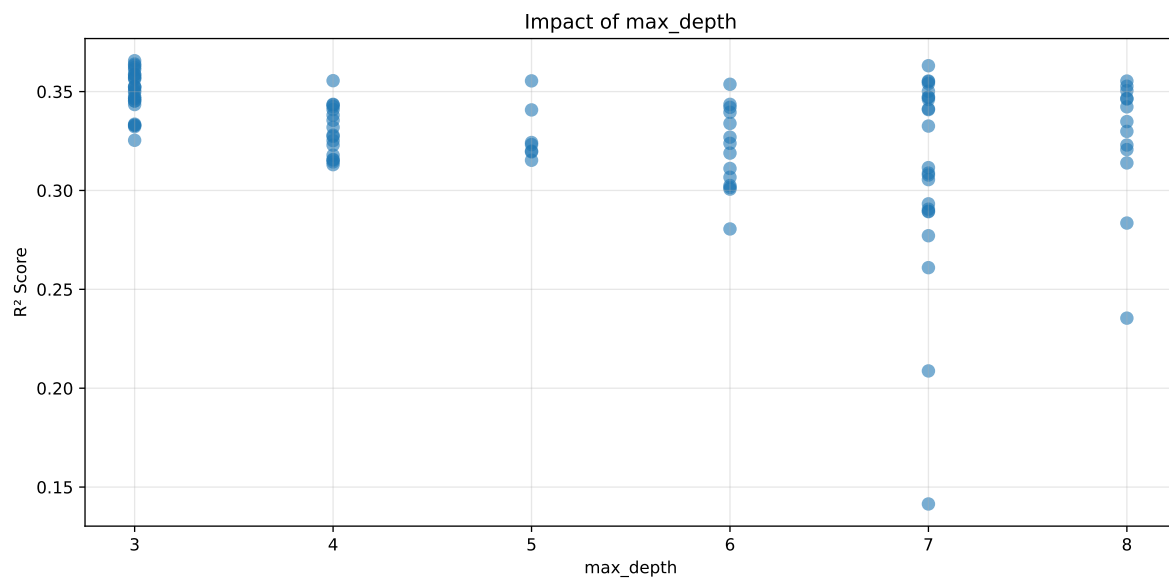
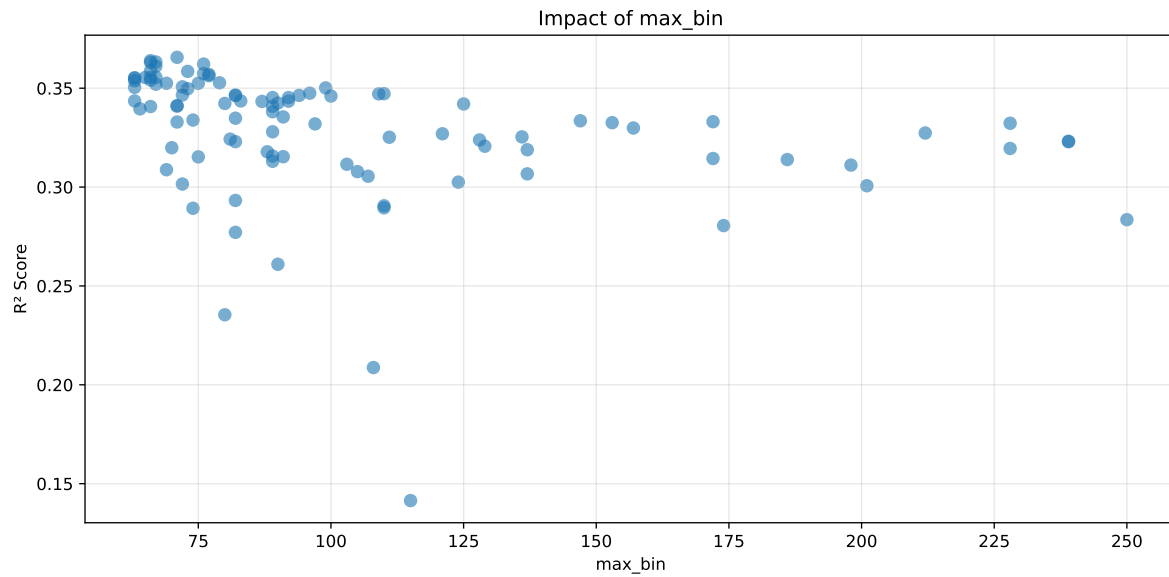
===== TOP Importance FEATURES =====

feature importance r_umidity 398 r_CO2 341 r_umidity_wl_24h_A_energy 340 r_NH3
300 r_THI 264 r_temperature_wl_24h_A_energy 239 r_temperature 228 r_umid-
ity_wl_24h_A_rms 228 r_temperature_wl_24h_A_rms 225 r_NH3_wl_24h_A_energy
195 r_umidity_wl_24h_D3_energy 195 r_temperature_wl_24h_D3_energy_rel 178
r_NH3_wl_24h_D2_energy_rel 173 r_temperature_wl_24h_D2_energy_rel 136 r_temper-
ature_wl_24h_D1_energy 123



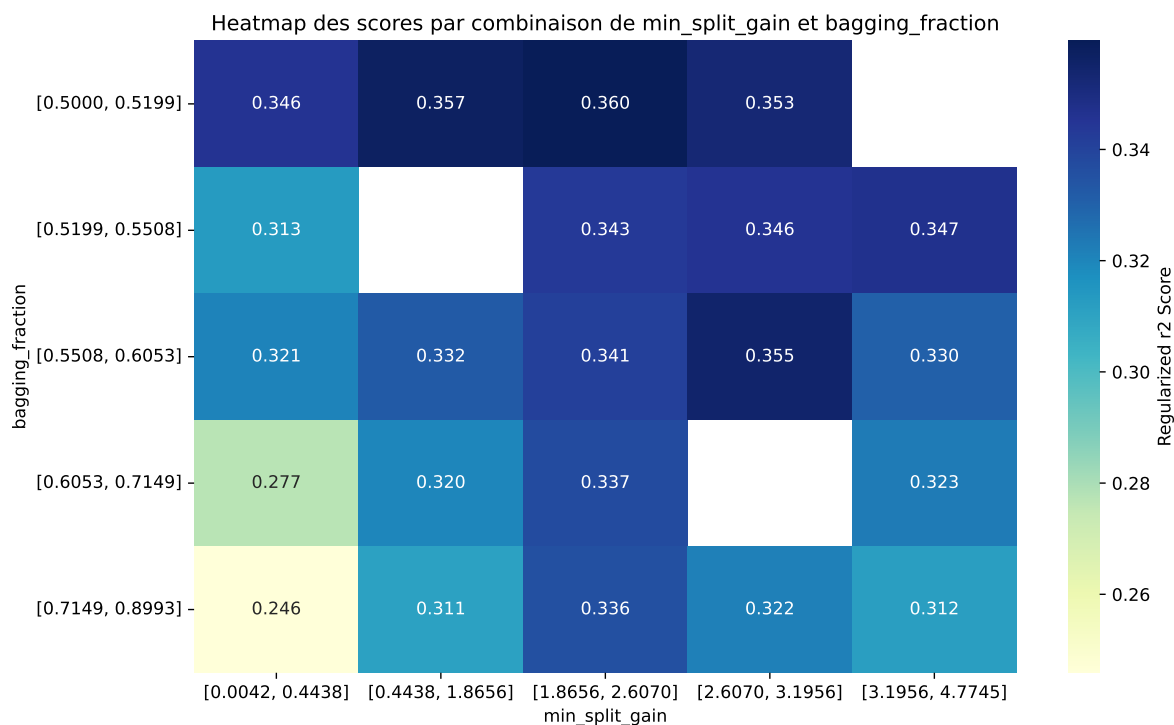
Paramètres avec >5% d'importance : ['min_split_gain', 'bagging_fraction', 'max_bin', 'max_depth']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6336	0.5889	0.5396	0.0092	0.3656	0.5646	-0.0243	0.9587	1.0758
2	0.6241	0.5808	0.5291	0.0055	0.3639	0.5541	-0.0267	0.9541	1.0747
3	0.5278	0.5004	0.4474	0.008	0.3633	0.4844	-0.016	0.9681	1.0548
4	0.5241	0.4973	0.4417	0.0109	0.3631	0.4763	-0.021	0.9579	1.054
5	0.5384	0.509	0.4531	0.0089	0.3622	0.486	-0.023	0.9548	1.0577
6	0.5318	0.5033	0.448	0.0138	0.3611	0.4863	-0.0171	0.9661	1.0565
7	0.6115	0.5694	0.517	0.0099	0.3588	0.5427	-0.0267	0.9531	1.074
8	0.6473	0.5991	0.547	0.0079	0.3585	0.5639	-0.0352	0.9412	1.0803
9	0.5491	0.5171	0.4616	0.0112	0.3574	0.4977	-0.0194	0.9625	1.0618
10	0.5772	0.5405	0.4871	0.0172	0.357	0.5233	-0.0172	0.9681	1.0679
---	---	---	---	---	---	---	---	---	---
100	0.8317	0.7167	0.6641	0.0094	0.1415	0.6944	-0.0223	0.9688	1.1605

Hyperparameters:

Rank 1: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.01999830122683536, 'reg__num_leaves': 43, 'reg__subsample': 0.7513911746384387, 'reg__colsample_bytree': 0.6432832200578964, 'reg__min_child_samples': 49, 'reg__reg_alpha': 9.863482815691578, 'reg__reg_lambda': 47.1068718818246, 'reg__min_split_gain': 0.07588461479178177, 'reg__path_smooth': 0.31122846390222275, 'reg__min_child_weight':

0.07208495998945501, 'reg__feature_fraction': 0.5018544313603263, 'reg__bagging_fraction': 0.501533493994242, 'reg__bagging_freq': 4, 'reg__max_bin': 71, 'reg__min_data_per_group': 146}

Rank 2: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.01815163743861651, 'reg__num_leaves': 42, 'reg__subsample': 0.7362690314569901, 'reg__colsample_bytree': 0.6177920413769785, 'reg__min_child_samples': 46, 'reg__reg_alpha': 9.933130882644313, 'reg__reg_lambda': 48.95684821698005, 'reg__min_split_gain': 0.5955578750314438, 'reg__path_smooth': 0.2834553375513984, 'reg__min_child_weight': 0.08226124571959795, 'reg__feature_fraction': 0.8771342693096791, 'reg__bagging_fraction': 0.6755897473335817, 'reg__bagging_freq': 5, 'reg__max_bin': 66, 'reg__min_data_per_group': 150}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.06187207773211148, 'reg__num_leaves': 43, 'reg__subsample': 0.683288545181963, 'reg__colsample_bytree': 0.6203889816000996, 'reg__min_child_samples': 29, 'reg__reg_alpha': 9.602005845478288, 'reg__reg_lambda': 24.332428985807976, 'reg__min_split_gain': 2.612798513878334, 'reg__path_smooth': 0.4868636294189782, 'reg__min_child_weight': 0.056456910510558296, 'reg__feature_fraction': 0.5645160228505468, 'reg__bagging_fraction': 0.5066389342200865, 'reg__bagging_freq': 2, 'reg__max_bin': 67, 'reg__min_data_per_group': 127}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.01840038551641314, 'reg__num_leaves': 45, 'reg__subsample': 0.7356572109692718, 'reg__colsample_bytree': 0.6305222427804826, 'reg__min_child_samples': 45, 'reg__reg_alpha': 9.953909198126794, 'reg__reg_lambda': 49.64659558779725, 'reg__min_split_gain': 2.7255825726601888, 'reg__path_smooth': 4.979953853428553, 'reg__min_child_weight': 0.1378957346106601, 'reg__feature_fraction': 0.8935161425742372, 'reg__bagging_fraction': 0.5035557050691636, 'reg__bagging_freq': 4, 'reg__max_bin': 66, 'reg__min_data_per_group': 144}

Rank 5: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.013234488165131921, 'reg__num_leaves': 43, 'reg__subsample': 0.6955254391946789, 'reg__colsample_bytree': 0.6172382799576001, 'reg__min_child_samples': 28, 'reg__reg_alpha': 9.613398271484174, 'reg__reg_lambda': 28.675148458625916, 'reg__min_split_gain': 2.605154858008136, 'reg__path_smooth': 0.3737692768625089, 'reg__min_child_weight': 0.19308912535383913, 'reg__feature_fraction': 0.8109410126788066, 'reg__bagging_fraction': 0.5070824687869235, 'reg__bagging_freq': 3, 'reg__max_bin': 76, 'reg__min_data_per_group': 126}

Rank 6: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.01231845551114674, 'reg__num_leaves': 44, 'reg__subsample': 0.722080268217373, 'reg__colsample_bytree': 0.6182218517174743, 'reg__min_child_samples': 36, 'reg__reg_alpha': 9.578064699884605, 'reg__reg_lambda': 25.499184948182922, 'reg__min_split_gain': 2.599299033777698, 'reg__path_smooth': 0.8435671775289979, 'reg__min_child_weight':

0.5771948614573307, 'reg__feature_fraction': 0.8039968506353125, 'reg__bagging_fraction': 0.5082202330967482, 'reg__bagging_freq': 3, 'reg__max_bin': 67, 'reg__min_data_per_group': 127}

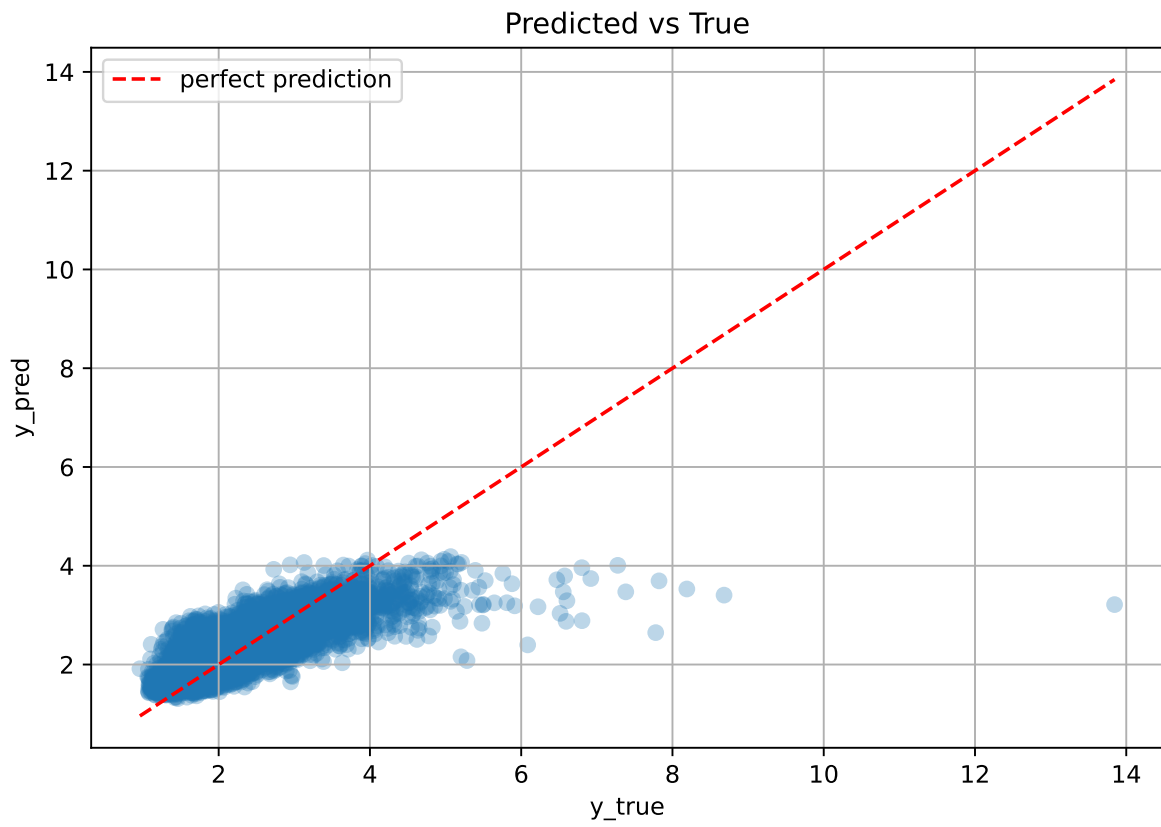
Rank 7: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.015670208847168875, 'reg__num_leaves': 44, 'reg__subsample': 0.7140076477295132, 'reg__colsample_bytree': 0.6218740066456324, 'reg__min_child_samples': 28, 'reg__reg_alpha': 4.346387689064639, 'reg__reg_lambda': 23.767518764754783, 'reg__min_split_gain': 1.3740496703018867, 'reg__path_smooth': 0.6785978522203704, 'reg__min_child_weight': 0.19840686239936062, 'reg__feature_fraction': 0.8086758048108034, 'reg__bagging_fraction': 0.5514612328429078, 'reg__bagging_freq': 3, 'reg__max_bin': 66, 'reg__min_data_per_group': 144}

Rank 8: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.012507222071709654, 'reg__num_leaves': 33, 'reg__subsample': 0.8602134359199916, 'reg__colsample_bytree': 0.6352459406733683, 'reg__min_child_samples': 34, 'reg__reg_alpha': 7.554859098945899, 'reg__reg_lambda': 47.19316918500649, 'reg__min_split_gain': 0.049872543618395326, 'reg__path_smooth': 0.5458199999537945, 'reg__min_child_weight': 0.2460421921842007, 'reg__feature_fraction': 0.7873912468079558, 'reg__bagging_fraction': 0.579897332735031, 'reg__bagging_freq': 4, 'reg__max_bin': 73, 'reg__min_data_per_group': 116}

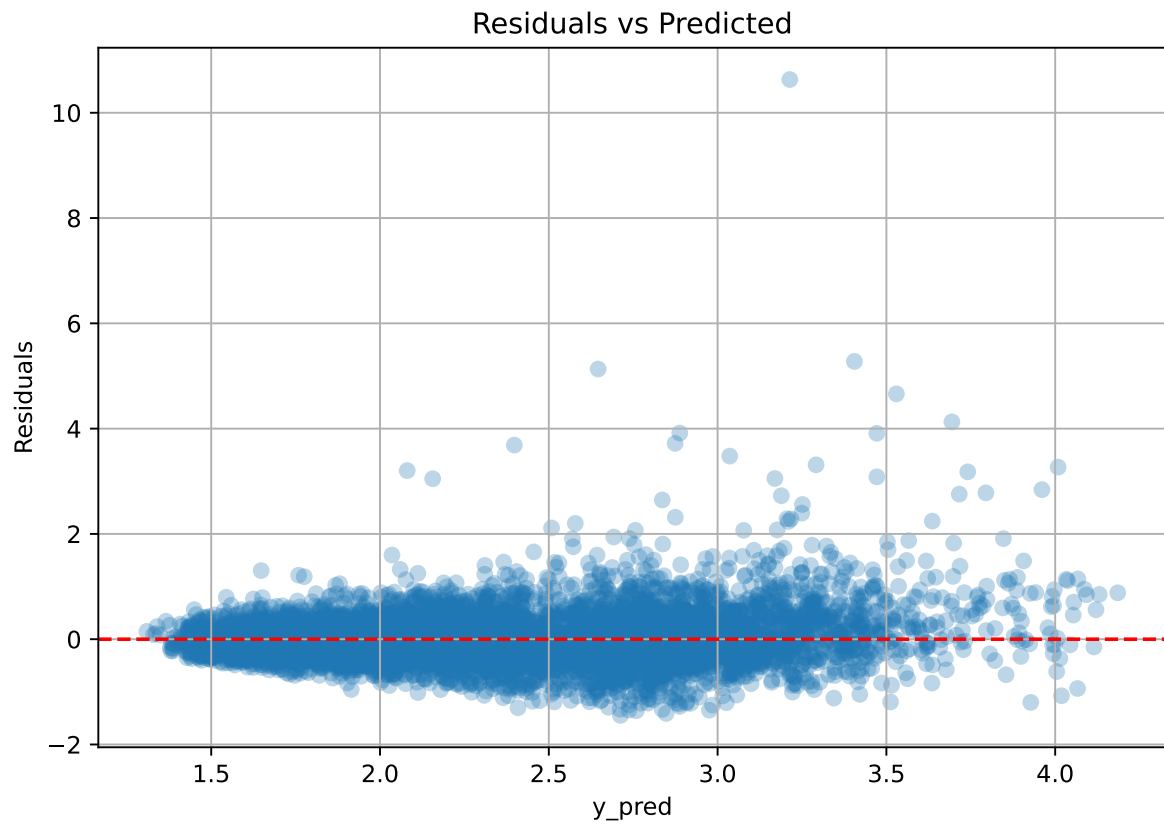
Rank 9: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.032313164477143504, 'reg__num_leaves': 44, 'reg__subsample': 0.7156537794297181, 'reg__colsample_bytree': 0.6204902276923963, 'reg__min_child_samples': 30, 'reg__reg_alpha': 8.439936406278102, 'reg__reg_lambda': 29.387380968279622, 'reg__min_split_gain': 2.6277798415023117, 'reg__path_smooth': 0.8451685385077317, 'reg__min_child_weight': 0.048918663614274764, 'reg__feature_fraction': 0.8835436516158012, 'reg__bagging_fraction': 0.5516730402653732, 'reg__bagging_freq': 2, 'reg__max_bin': 76, 'reg__min_data_per_group': 141}

Rank 10: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.013163445555146548, 'reg__num_leaves': 43, 'reg__subsample': 0.6822201440522614, 'reg__colsample_bytree': 0.618724790706376, 'reg__min_child_samples': 35, 'reg__reg_alpha': 9.648287496967995, 'reg__reg_lambda': 22.871795128607378, 'reg__min_split_gain': 1.3741460936111287, 'reg__path_smooth': 0.531751712842548, 'reg__min_child_weight': 0.05653016721172646, 'reg__feature_fraction': 0.8834110755896359, 'reg__bagging_fraction': 0.50894938913837, 'reg__bagging_freq': 3, 'reg__max_bin': 77, 'reg__min_data_per_group': 146}

3.5.2 Scatter



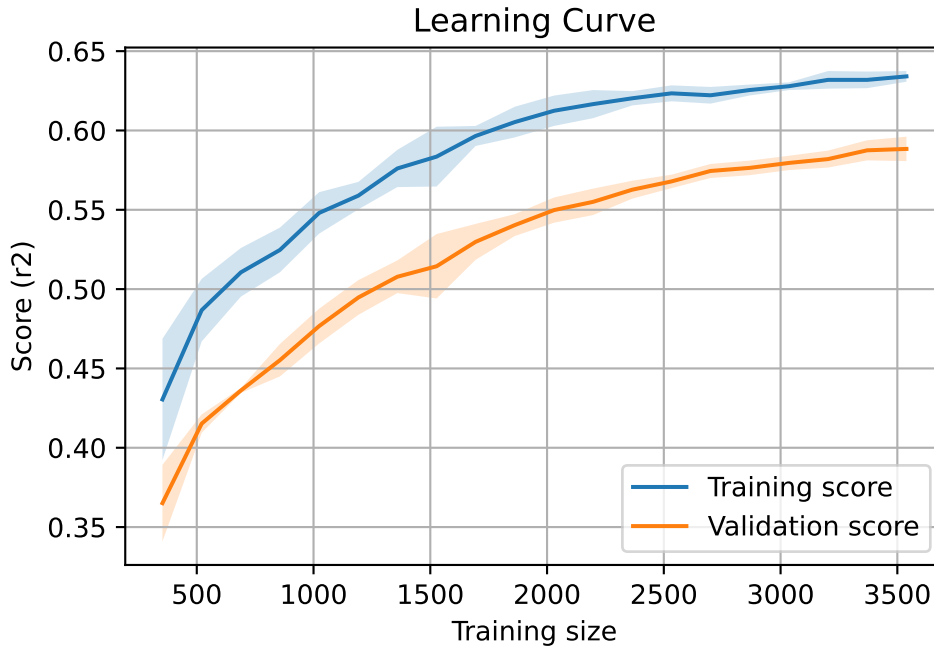
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7224, `Len(training)` : 5057,

`Len(training_real)` : 3540, `Len(test_of_training)` : 1517, `Len(test)` : 2167



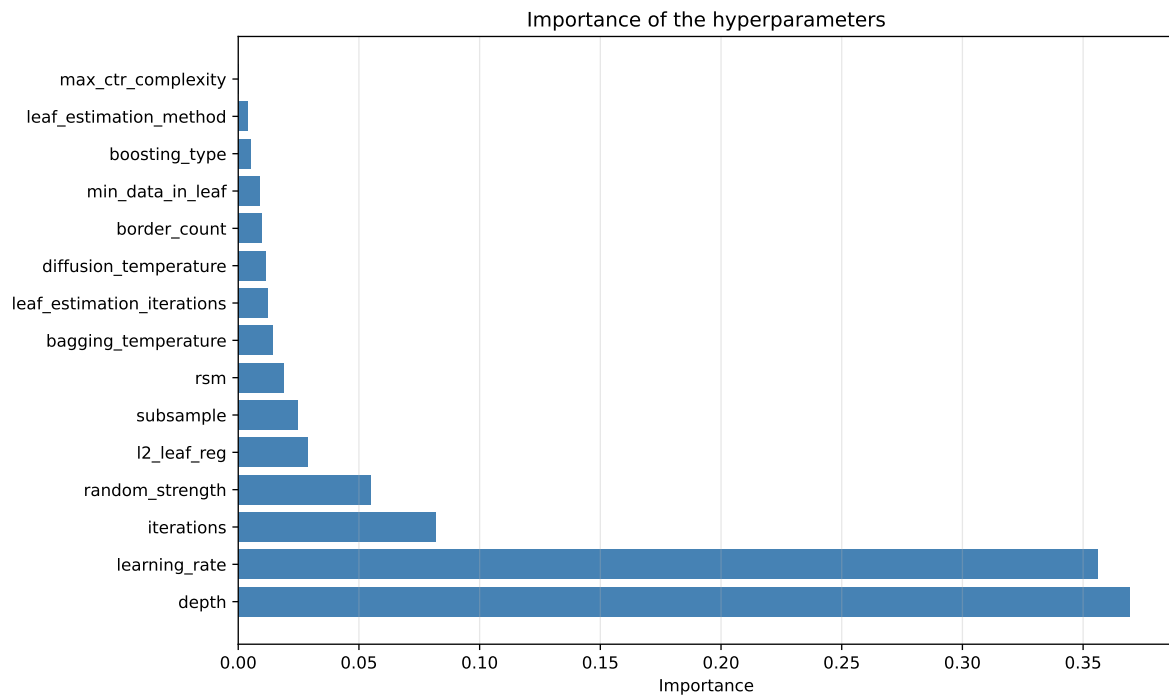
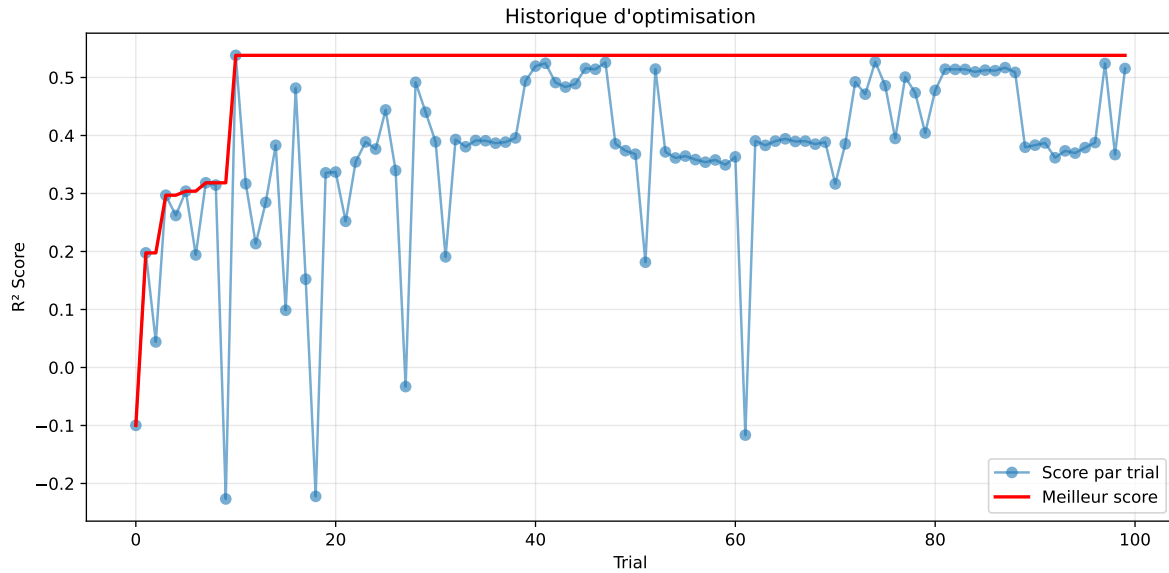
3.6 Model : CatBoost

3.6.1 Gridsearch

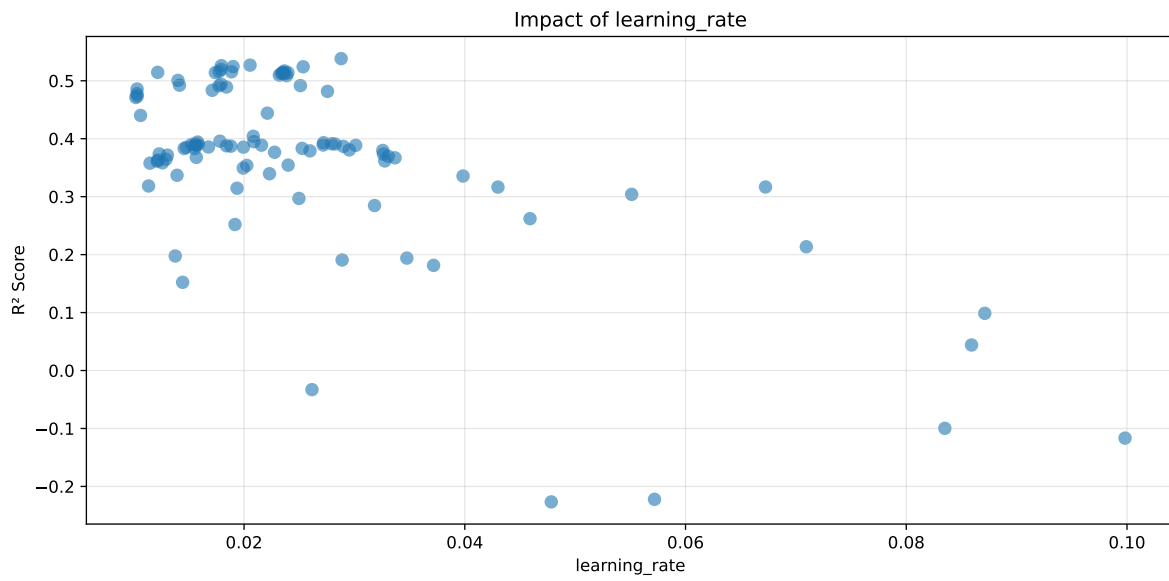
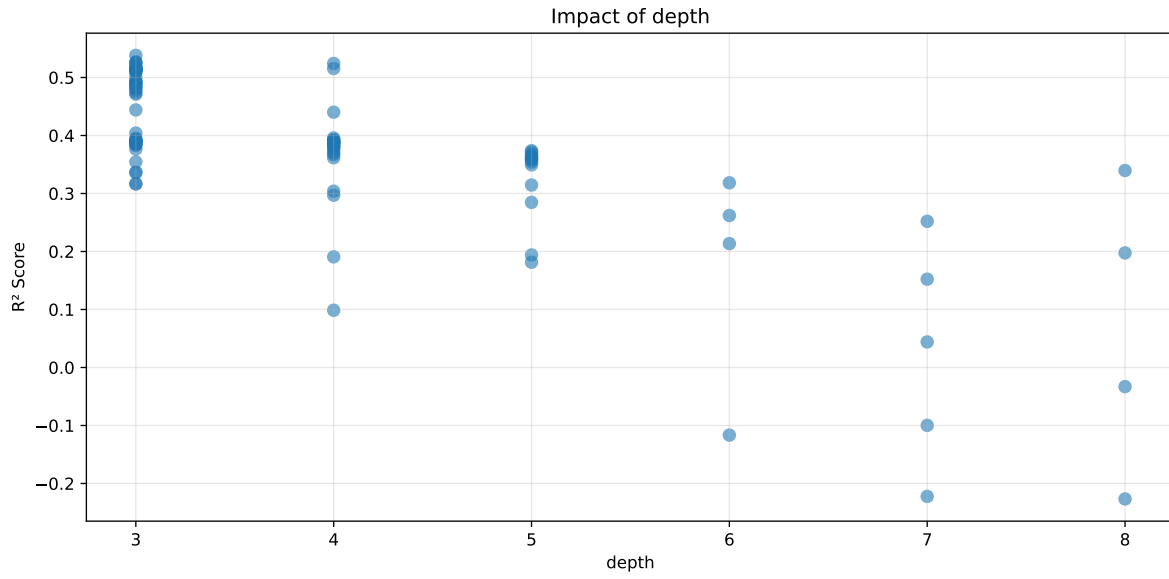
Distribution y_train vs y_test: y_train: mean=2.346, std=0.787, min=0.959, max=8.190
y_test: mean=2.382, std=0.828, min=1.071, max=13.845

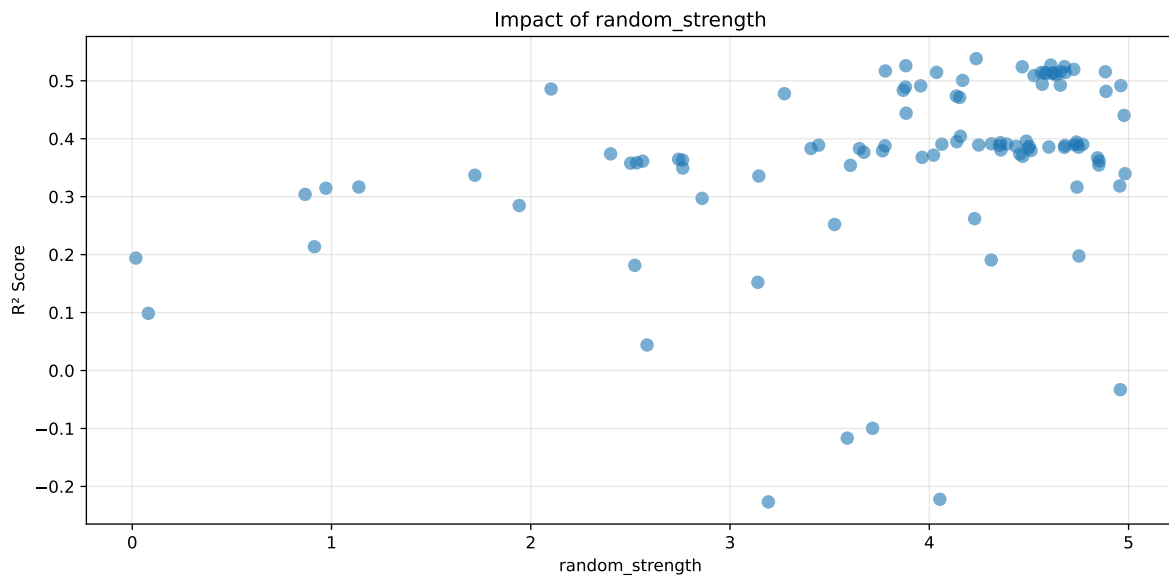
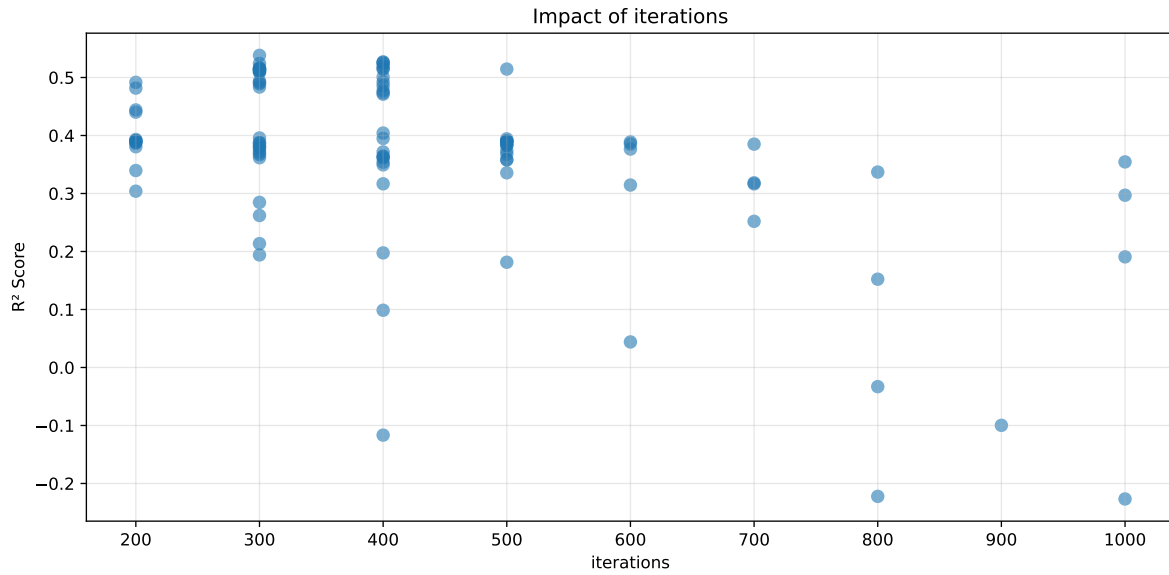
===== TOP Importance FEATURES =====

feature	importance
r_umidity	28.407079
r_temperature	15.545881
r_NH3	13.184350
r_umidity_wl_24h_A_energy	6.601384
r_CO2	5.354827
r_THI	4.419833
r_umidity_wl_24h_D3_energy	4.187143
r_temperature_wl_24h_A_rms	3.666942
r_temperature_wl_24h_A_energy	3.211560
r_NH3_wl_24h_A_energy	3.173641
r_umidity_wl_24h_A_rms	2.893848
r_NH3_wl_24h_D2_energy_rel	2.819019
r_temperature_wl_24h_D1_energy	2.613389
r_temperature_wl_24h_D3_energy_rel	2.433991
r_temperature_wl_24h_D2_energy_rel	1.487111



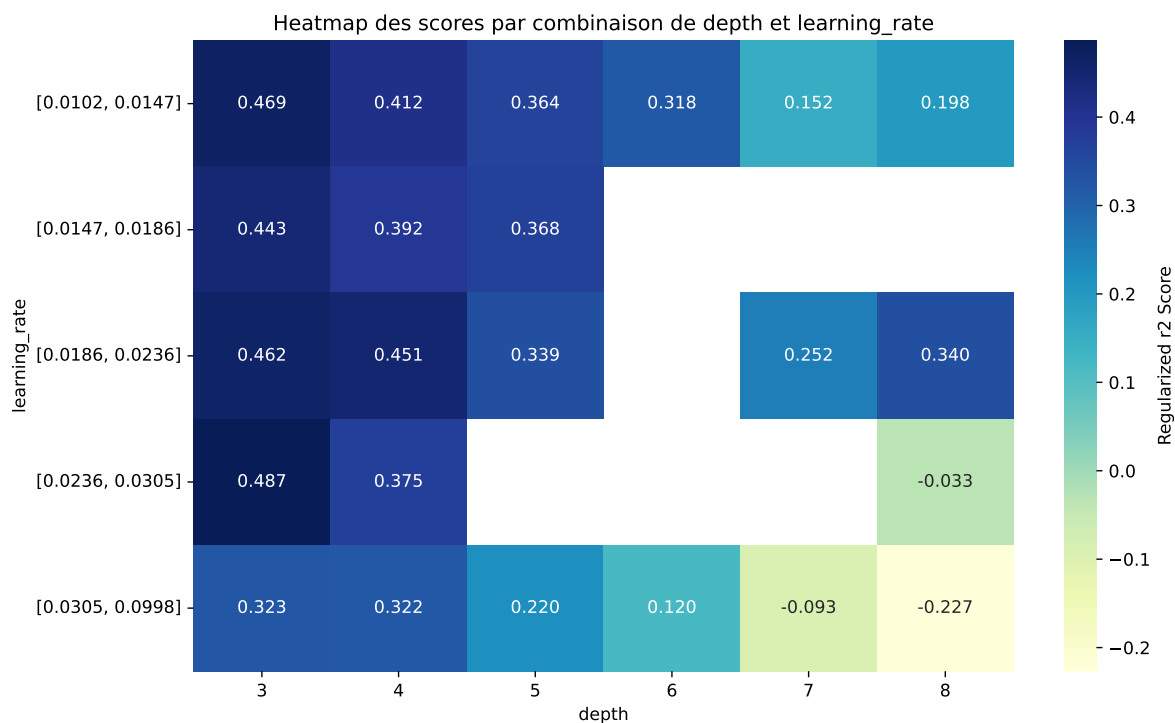
Paramètres avec >5% d'importance : ['depth', 'learning_rate', 'iterations', 'random_strength']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5624	0.5382	0.4858	0.0051	0.5382	0.4932	-0.045	0.9165	1.045
2	0.5527	0.527	0.4755	0.0047	0.527	0.4845	-0.0425	0.9194	1.0489
3	0.5521	0.526	0.4752	0.0044	0.526	0.4808	-0.0452	0.9141	1.0496
4	0.5497	0.5245	0.4741	0.0049	0.5245	0.4796	-0.0449	0.9144	1.048
5	0.5501	0.5242	0.4782	0.0062	0.5242	0.4854	-0.0388	0.926	1.0495
6	0.5444	0.5196	0.4687	0.0051	0.5196	0.4793	-0.0403	0.9224	1.0477
7	0.542	0.517	0.467	0.0054	0.517	0.4747	-0.0423	0.9182	1.0484
8	0.5398	0.5158	0.4662	0.0056	0.5158	0.4735	-0.0423	0.918	1.0465
9	0.5412	0.5155	0.4676	0.0064	0.5155	0.4688	-0.0467	0.9094	1.0498
10	0.5396	0.5146	0.4635	0.005	0.5146	0.4701	-0.0445	0.9135	1.0487
---	---	---	---	---	---	---	---	---	---
100	0.9769	0.7763	0.7374	0.0111	-0.2267	0.7606	-0.0157	0.9798	1.2584

Hyperparameters:

Rank 1: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.02879722867568162, 'reg__l2_leaf_reg': 19.30028137620615, 'reg__bagging_temperature': 1.5500586991411063, 'reg__random_strength': 4.23541487053455, 'reg__subsample': 0.6439440635686035, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.683345744440041, 'reg__border_count': 104, 'reg__leaf_estimation_method': 'Newton',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 789.2474305934438}

Rank 2: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.020541353868455636, 'reg__l2_leaf_reg': 30.591151994561486, 'reg__bagging_temperature': 0.8522990722881966, 'reg__random_strength': 4.609708776573604, 'reg__subsample': 0.710865521747452, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.6334475077754984, 'reg__border_count': 124, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1593.8487513409286}

Rank 3: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.017941485950288643, 'reg__l2_leaf_reg': 13.041905549702284, 'reg__bagging_temperature': 1.02839295304656, 'reg__random_strength': 3.882171764089678, 'reg__subsample': 0.6789491811745614, 'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.6296302417284733, 'reg__border_count': 128, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1020.3680339106357}

Rank 4: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.019002771856731576, 'reg__l2_leaf_reg': 26.563332310631328, 'reg__bagging_temperature': 0.9377833942556526, 'reg__random_strength': 4.6785881638554985, 'reg__subsample': 0.558115316967209, 'reg__min_data_in_leaf': 33, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.8358175541677725, 'reg__border_count': 124, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1998.3721215054115}

Rank 5: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.02535366736102054, 'reg__l2_leaf_reg': 27.087118005660294, 'reg__bagging_temperature': 1.231133542688456, 'reg__random_strength': 4.466546583584862, 'reg__subsample': 0.6271666736997455, 'reg__min_data_in_leaf': 12, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.8732804318525303, 'reg__border_count': 128, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 14.781845743683789}

Rank 6: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.017918562637329168, 'reg__l2_leaf_reg': 26.206334814459822, 'reg__bagging_temperature': 1.0902853026824297, 'reg__random_strength': 4.725228248075472, 'reg__subsample': 0.6407175021146241, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.8383594099606755, 'reg__border_count': 128, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 2034.0309331277765}

Rank 7: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.023654789560593862, 'reg__l2_leaf_reg': 27.05420065537511, 'reg__bagging_temperature': 0.6487180819477399, 'reg__random_strength': 3.7793930242748375, 'reg__subsample': 0.6421471014514593,

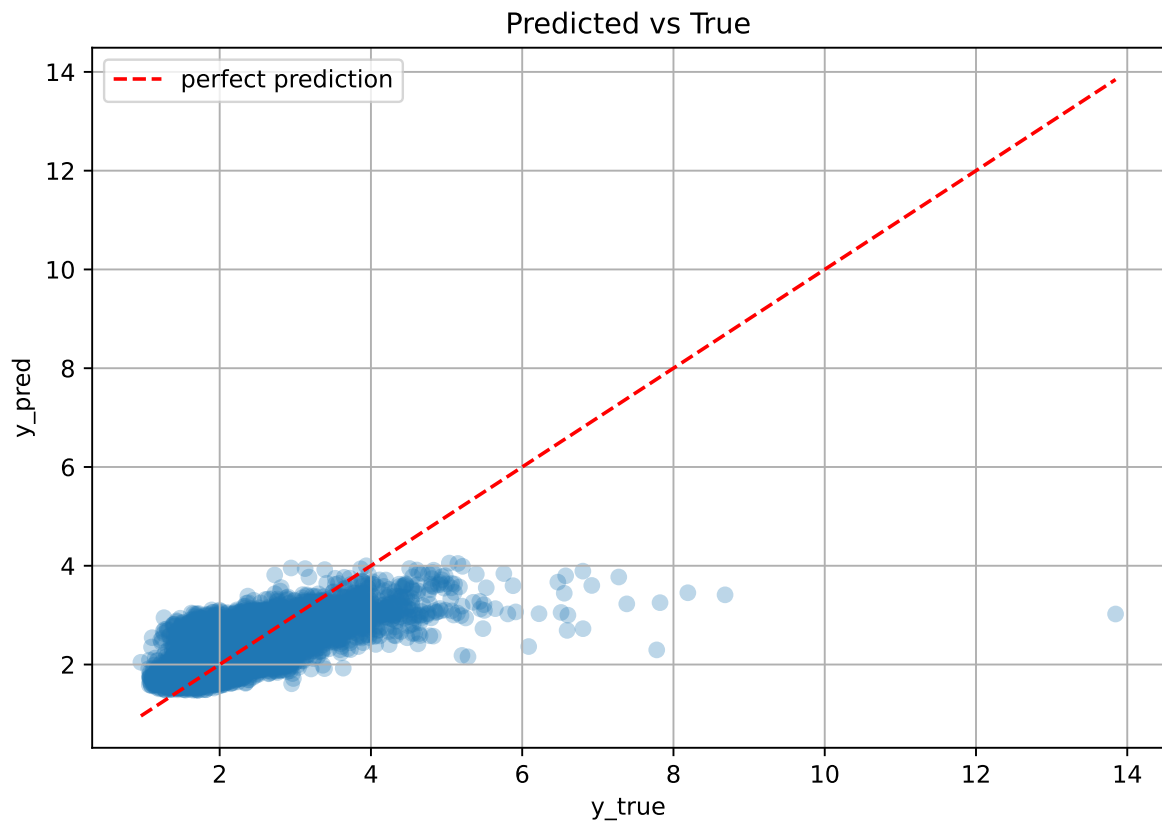
'reg__min_data_in_leaf': 12, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.5278248267958183, 'reg__border_count': 119, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 383.18826432479227}

Rank 8: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.0177337125051126, 'reg__l2_leaf_reg': 30.098054394928095, 'reg__bagging_temperature': 1.0318468059299952, 'reg__random_strength': 4.657972828687717, 'reg__subsample': 0.6747077208004912, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.6357237104380148, 'reg__border_count': 126, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1143.6053005772574}

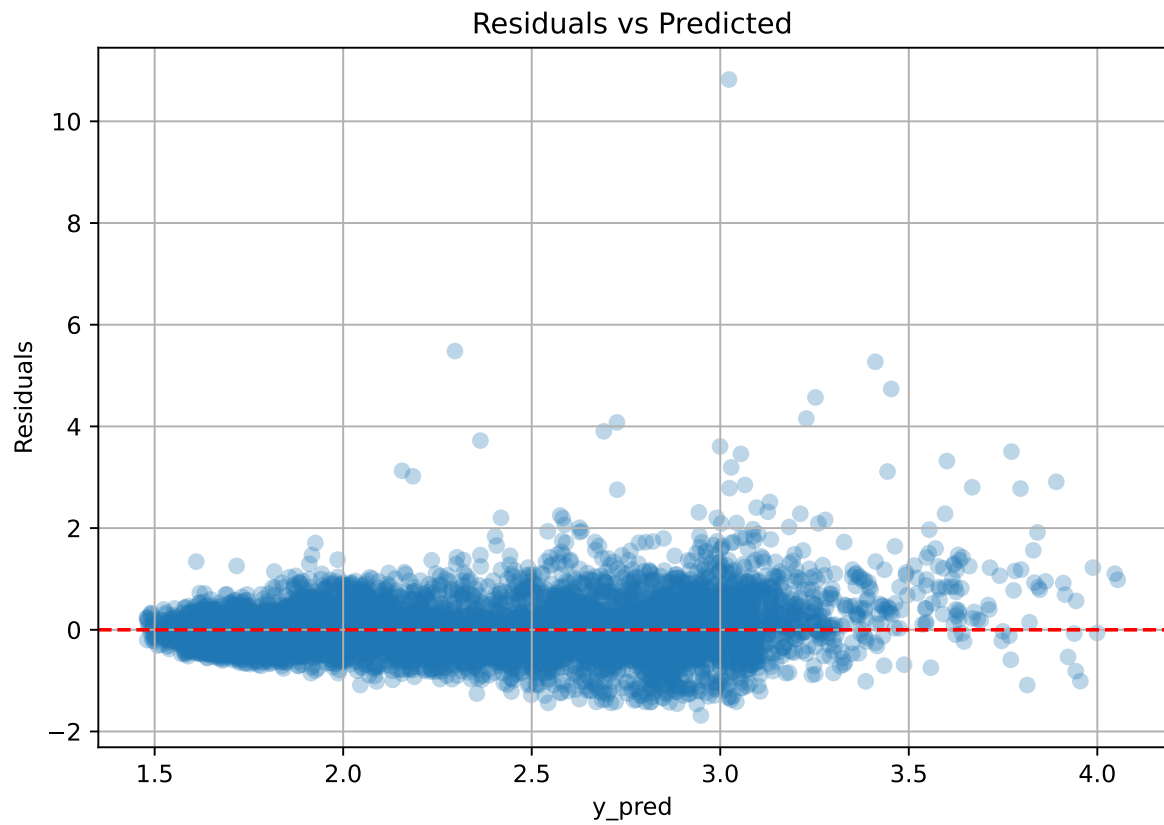
Rank 9: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.01886358306666488, 'reg__l2_leaf_reg': 23.494654184889168, 'reg__bagging_temperature': 0.19018323071769744, 'reg__random_strength': 4.8833607115404964, 'reg__subsample': 0.6459279935313288, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.5205723290005639, 'reg__border_count': 128, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 199.9153623336142}

Rank 10: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.012173011834398902, 'reg__l2_leaf_reg': 12.575755093367368, 'reg__bagging_temperature': 0.4637542658545645, 'reg__random_strength': 4.036394111662844, 'reg__subsample': 0.6774685700295757, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.6700378451625816, 'reg__border_count': 123, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 8802.704775270196}

3.6.2 Scatter



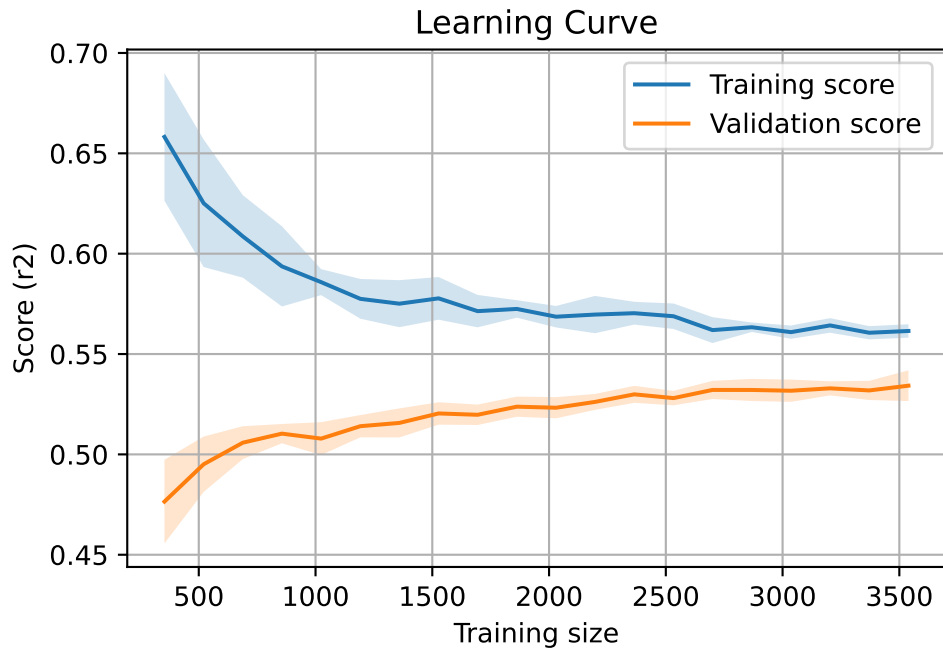
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7224, `Len(training)` : 5057,

`Len(training_real)` : 3540, `Len(test_of_training)` : 1517, `Len(test)` : 2167



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.422	0.413	0.008	0.387	0.389	-0.024	0.942	1.023	nan
LGBM	0.634	0.589	0.009	0.54	0.565	-0.024	0.959	1.076	0.3656
CatBoost	0.562	0.538	0.005	0.486	0.493	-0.045	0.916	1.045	0.5382

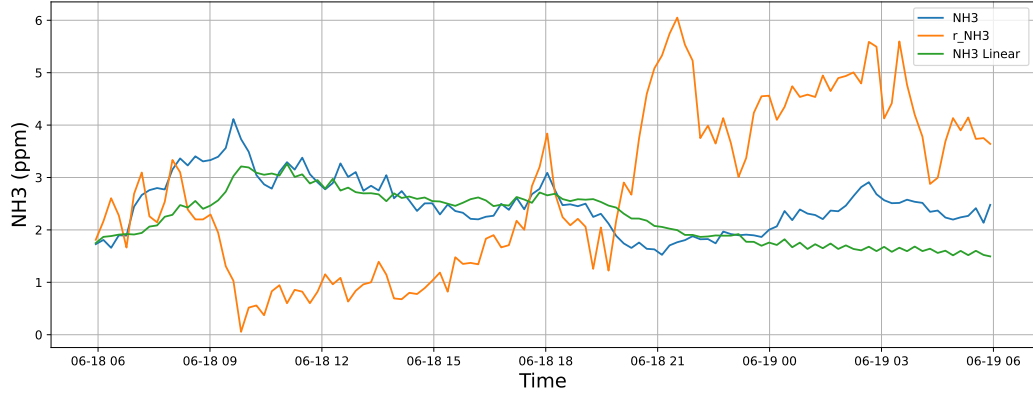
4 Plot

4.1 Standalone plots

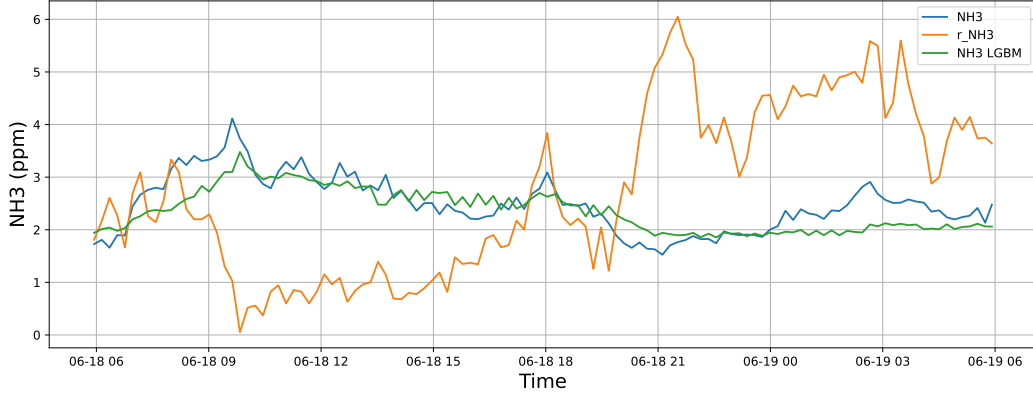
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

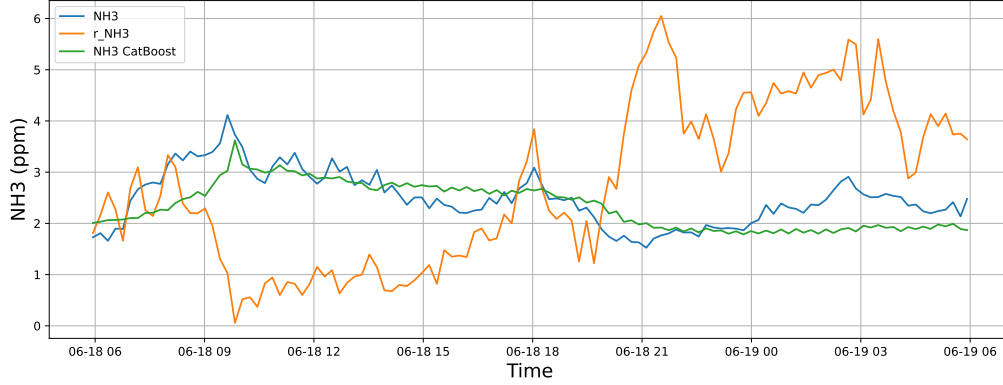
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

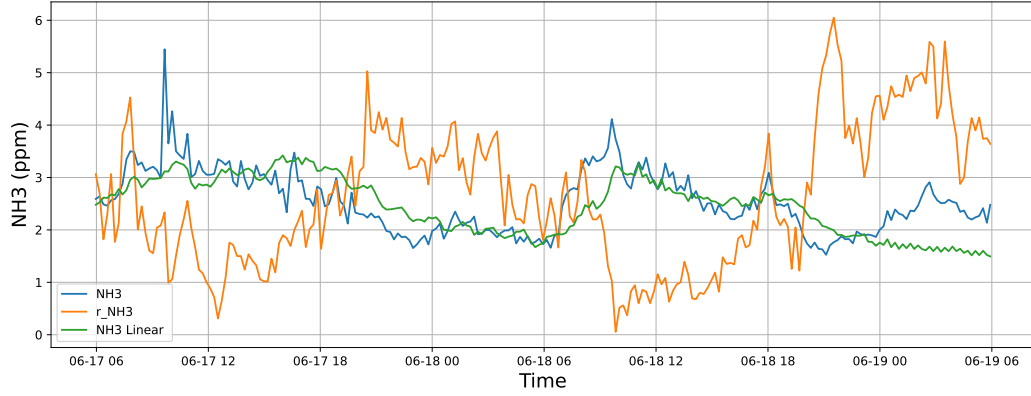


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

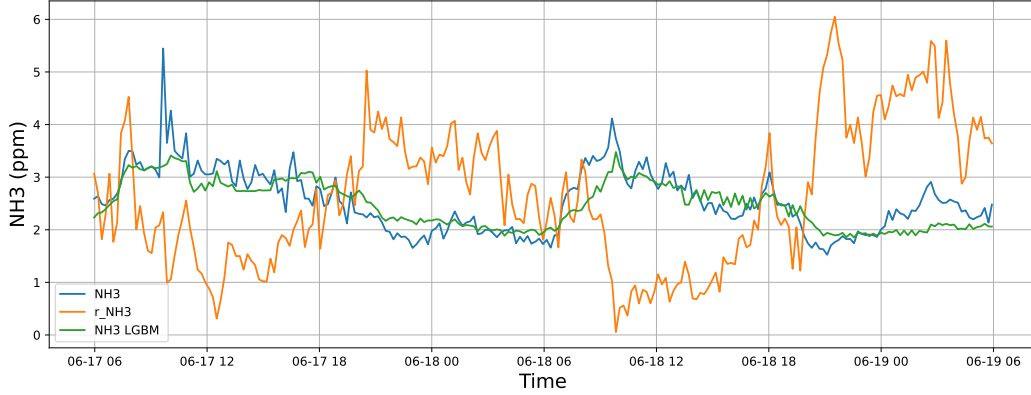


4.1.1.2 Time window : 48

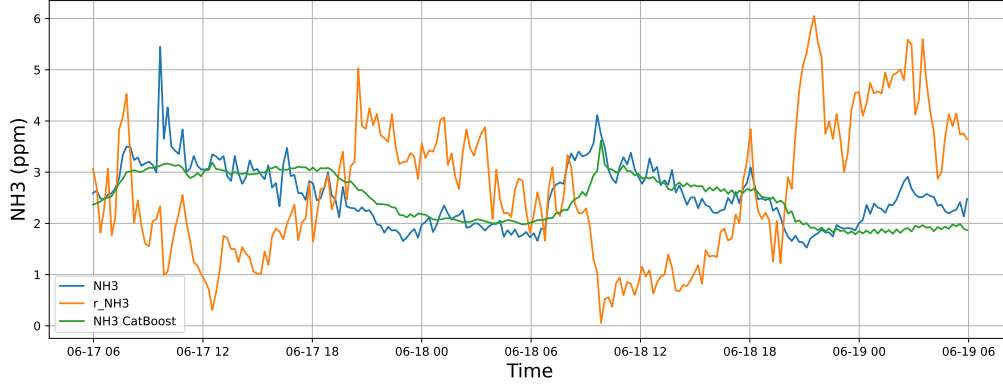
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

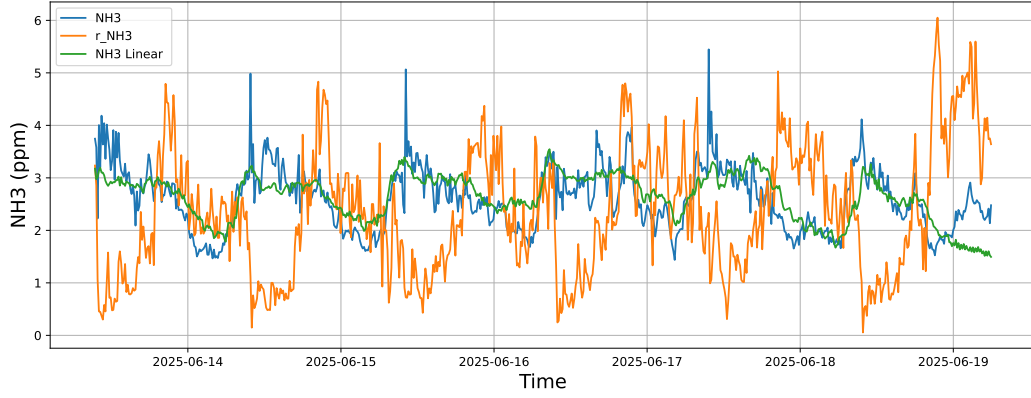


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

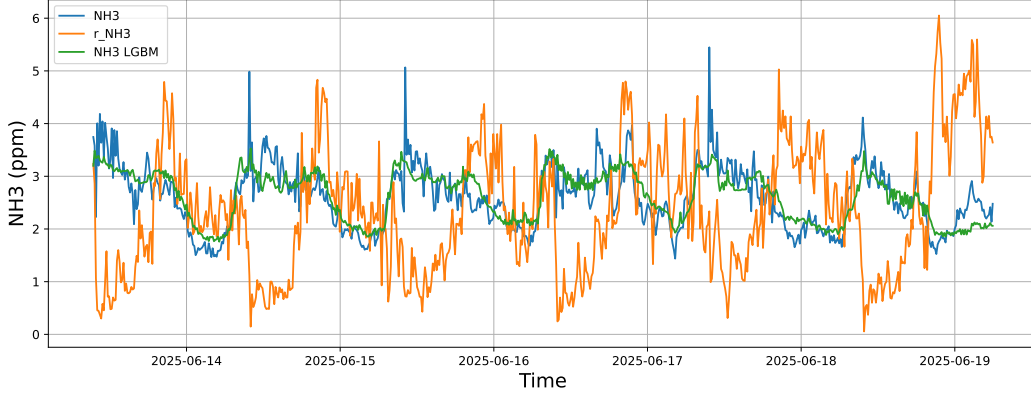


4.1.1.3 Time window : ALL

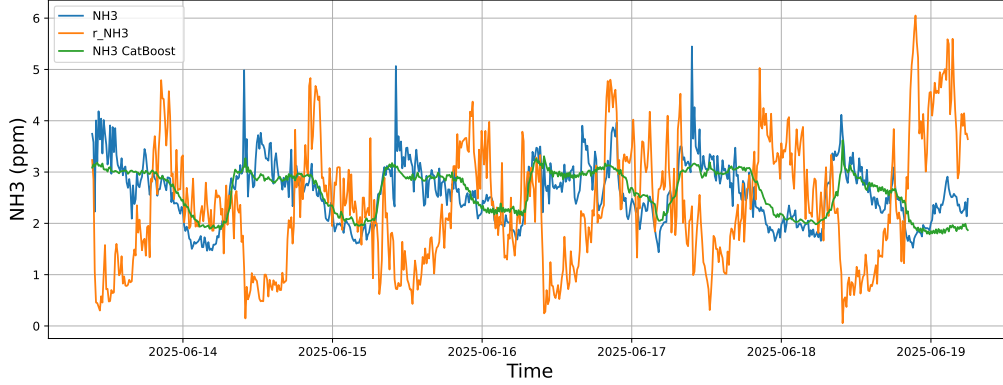
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



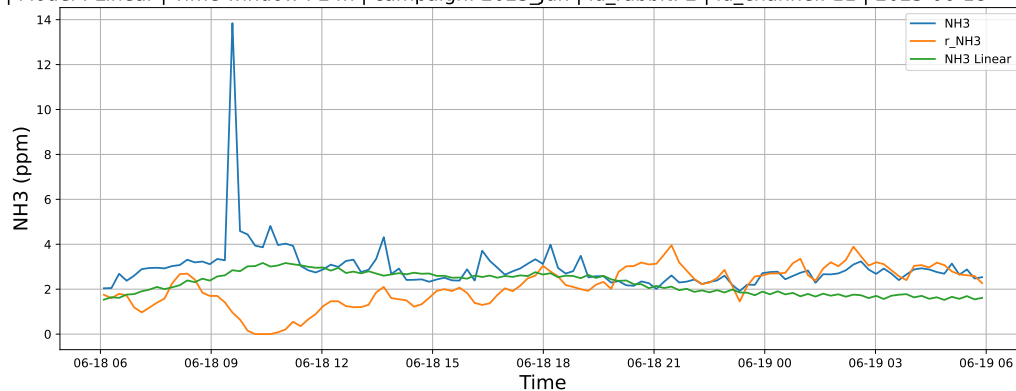
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



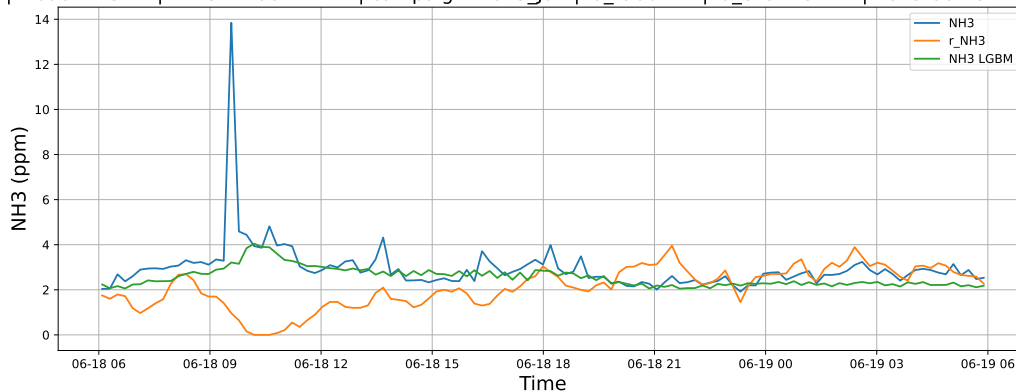
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

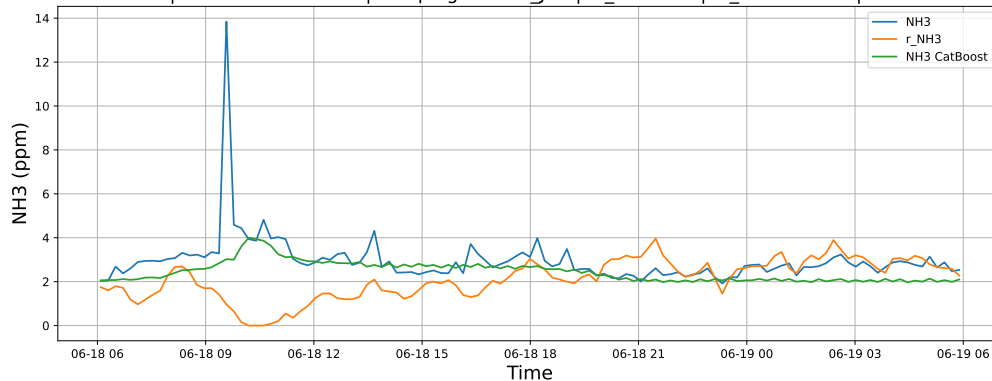
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

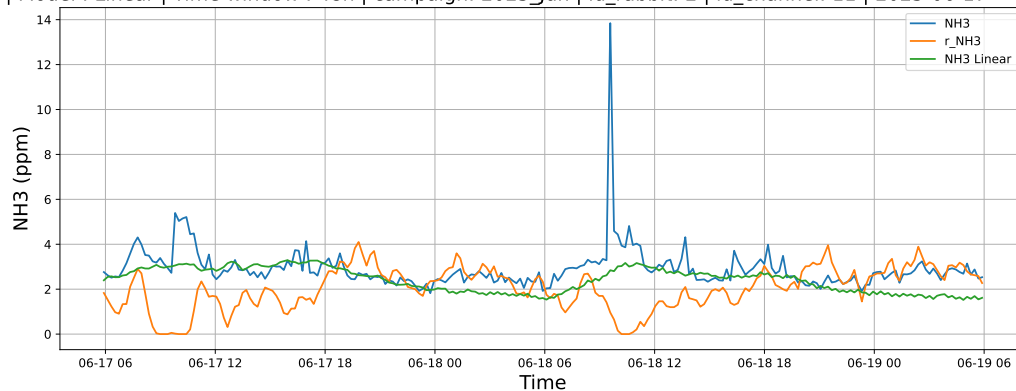


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

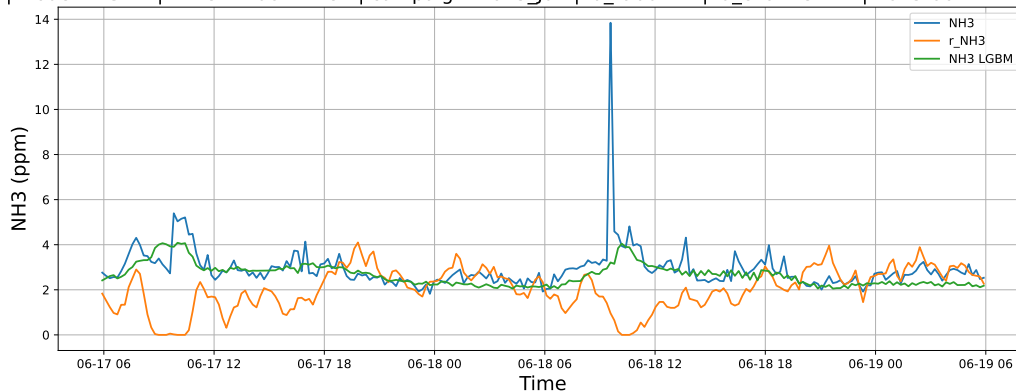


4.1.2.2 Time window : 48

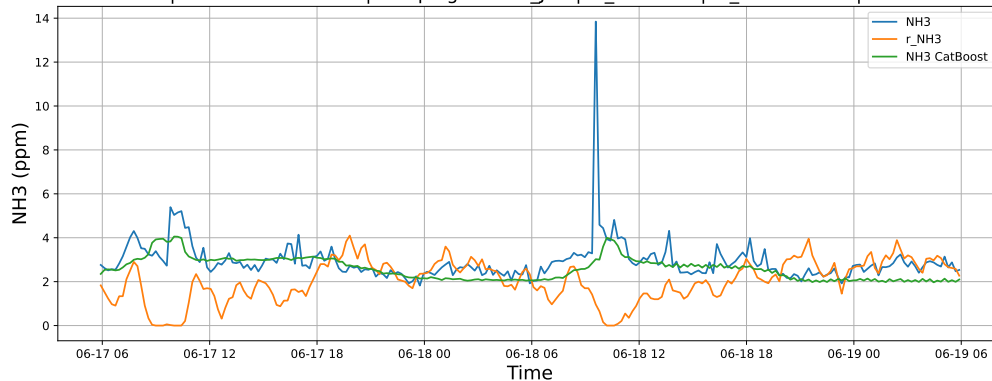
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

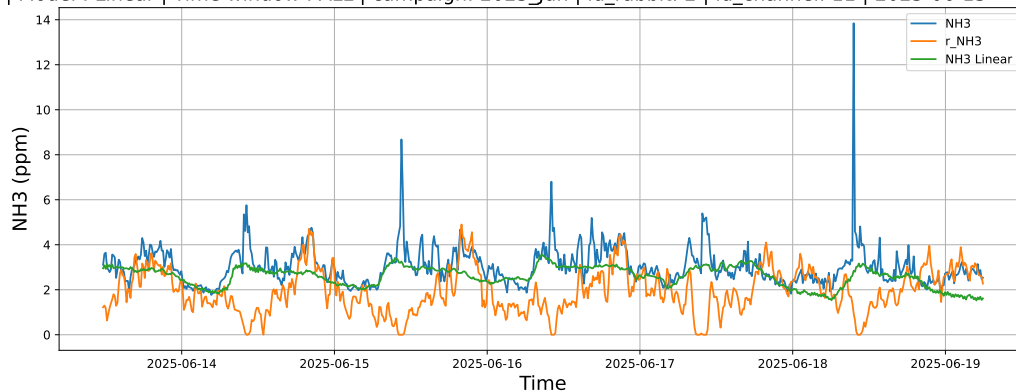


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

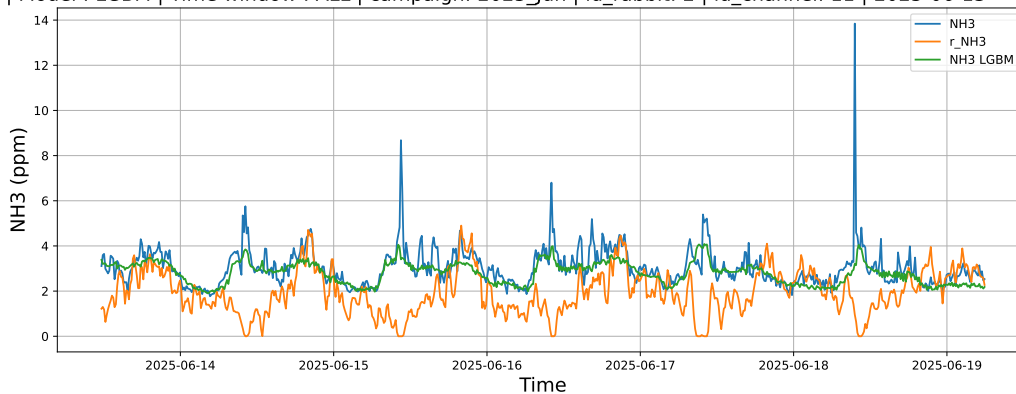


4.1.2.3 Time window : ALL

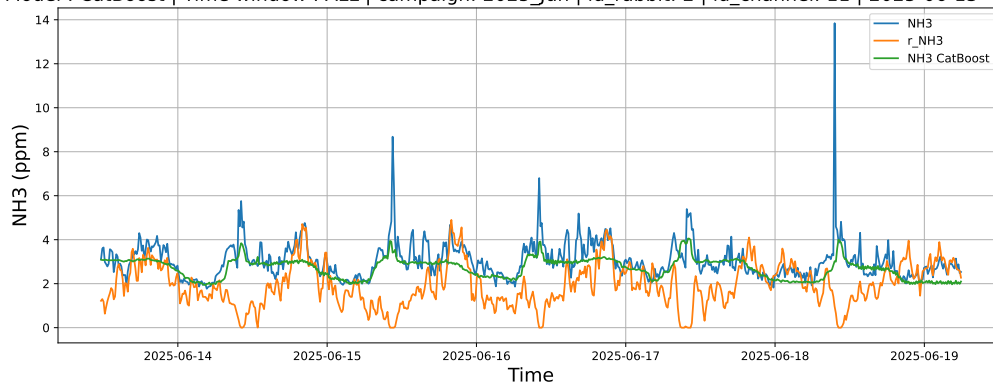
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



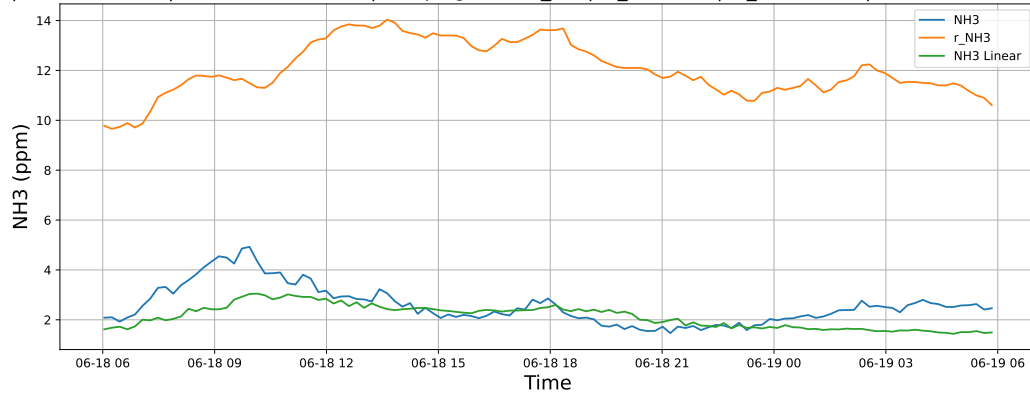
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



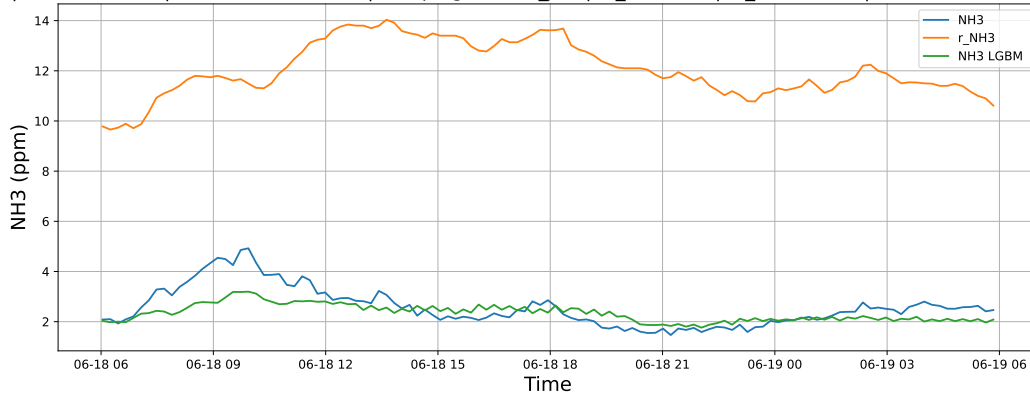
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

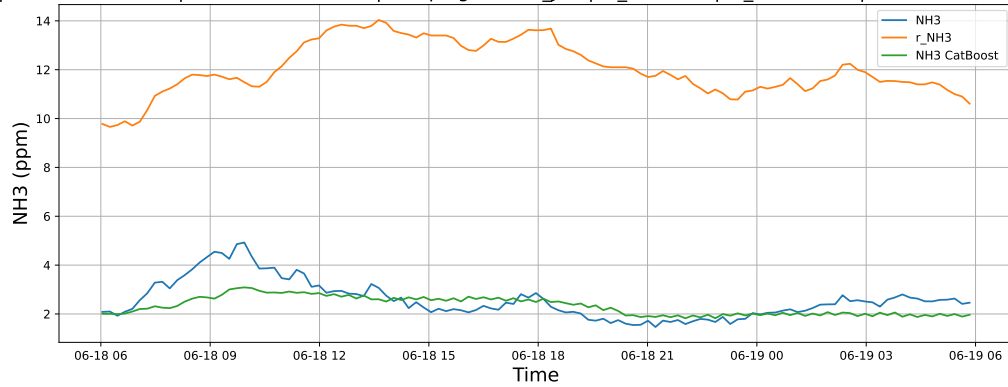
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

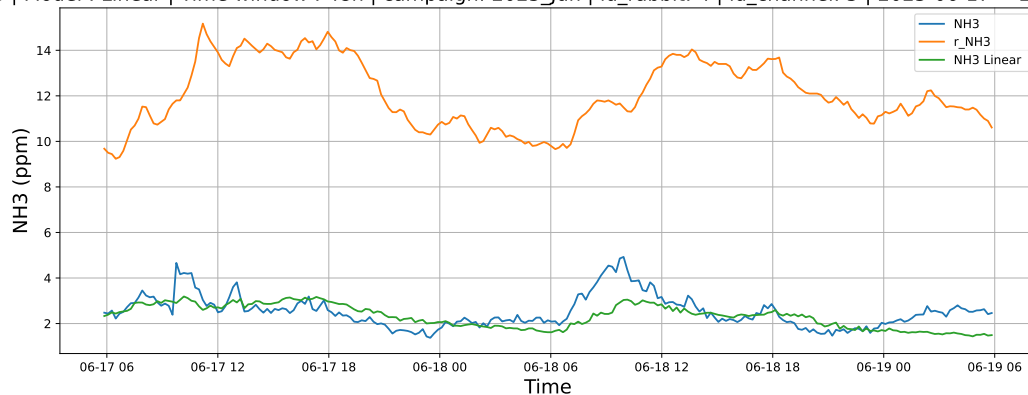


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

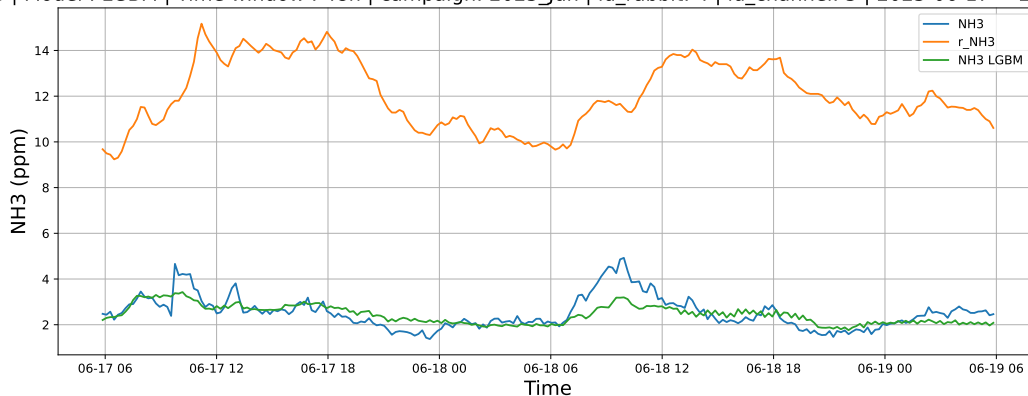


4.1.3.2 Time window : 48

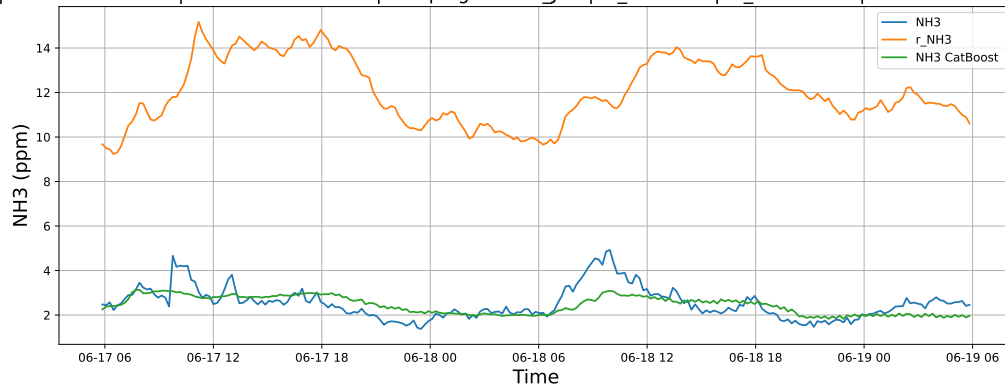
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

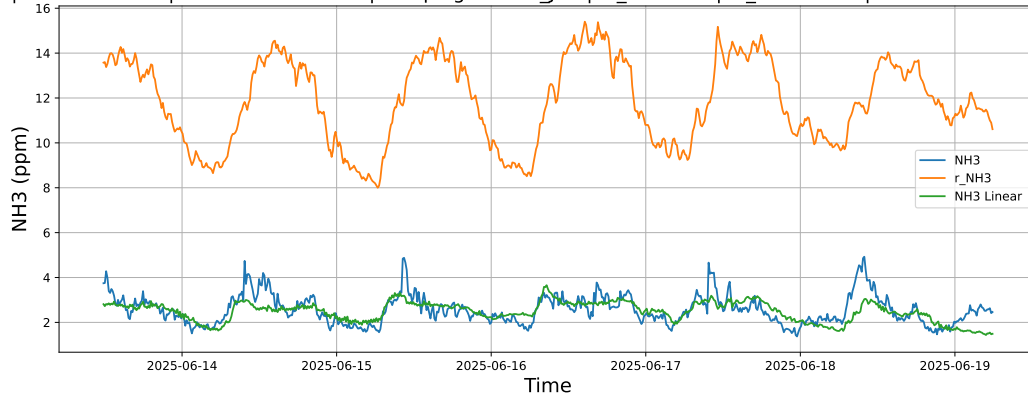


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

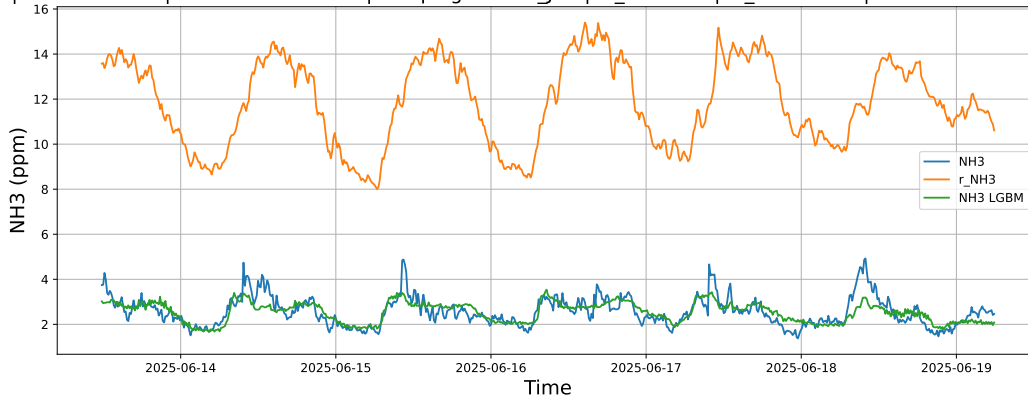


4.1.3.3 Time window : ALL

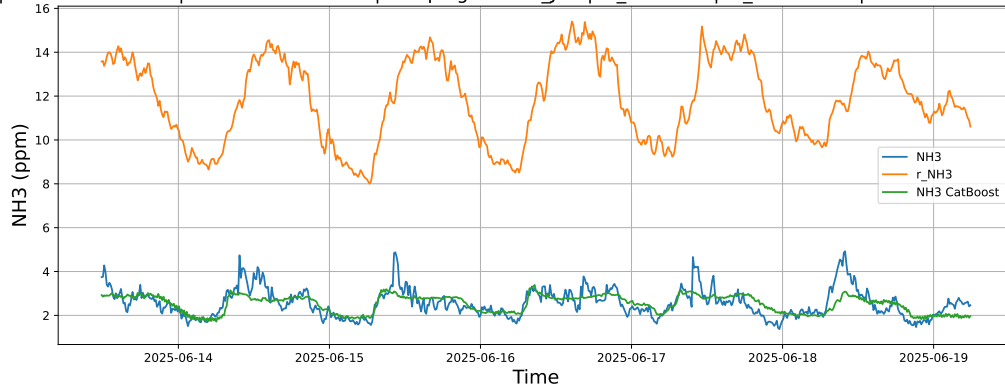
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



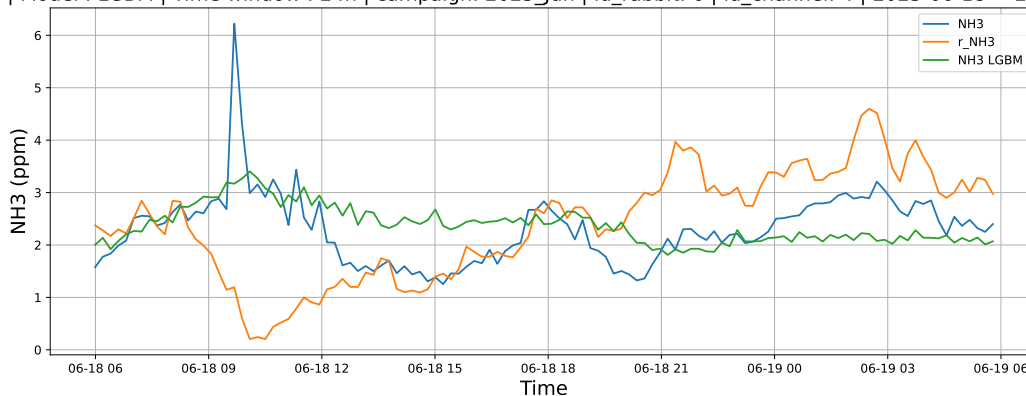
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

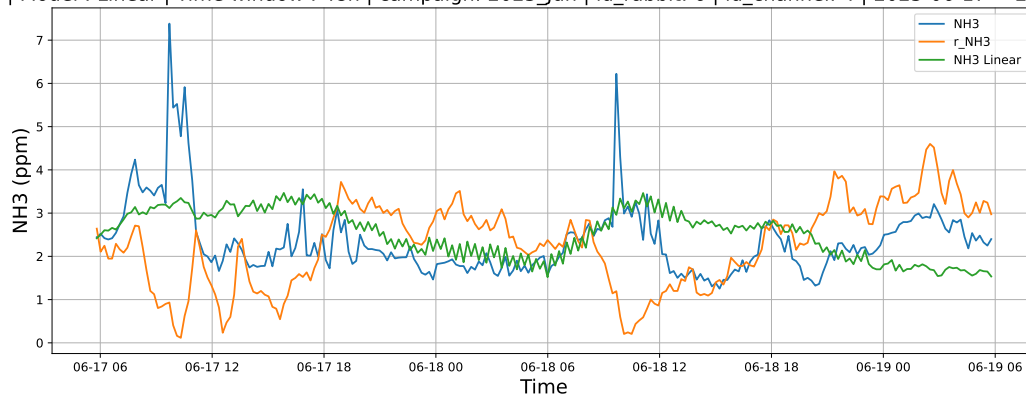


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

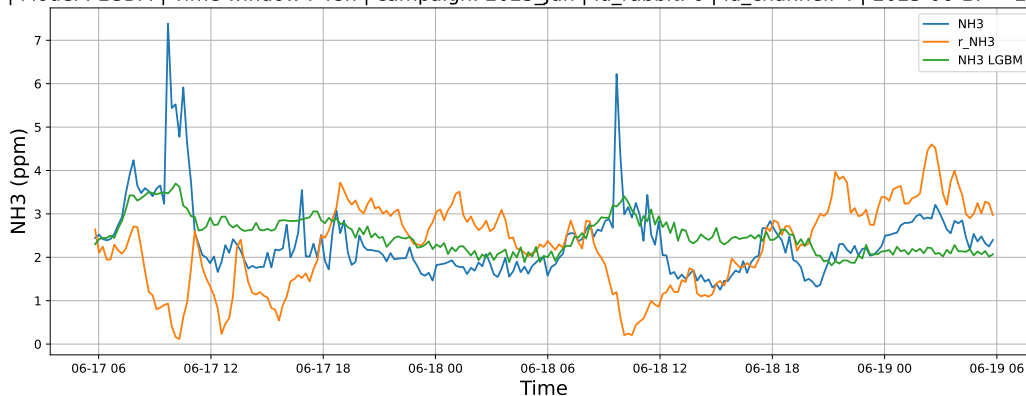


4.1.4.2 Time window : 48

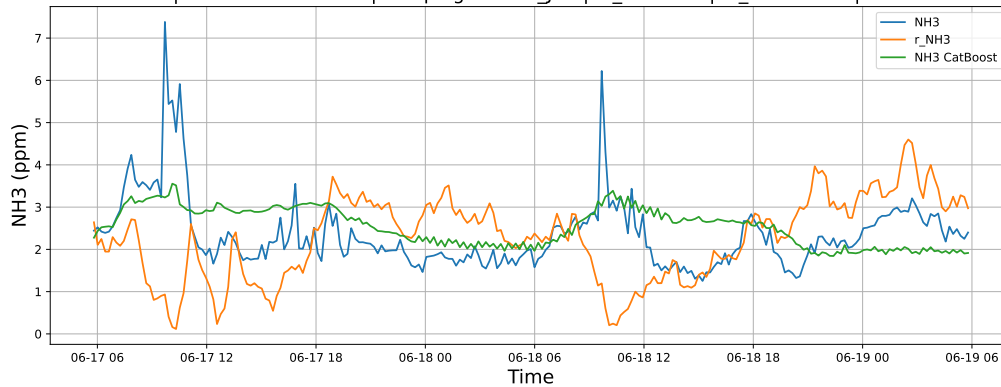
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

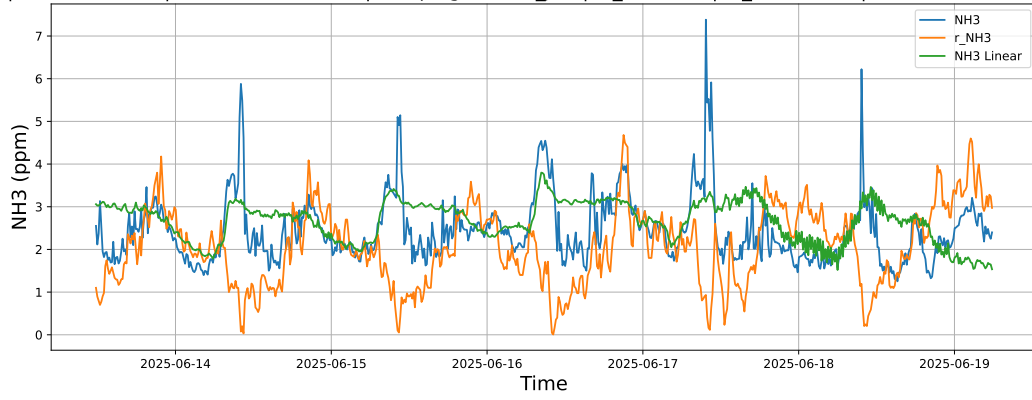


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

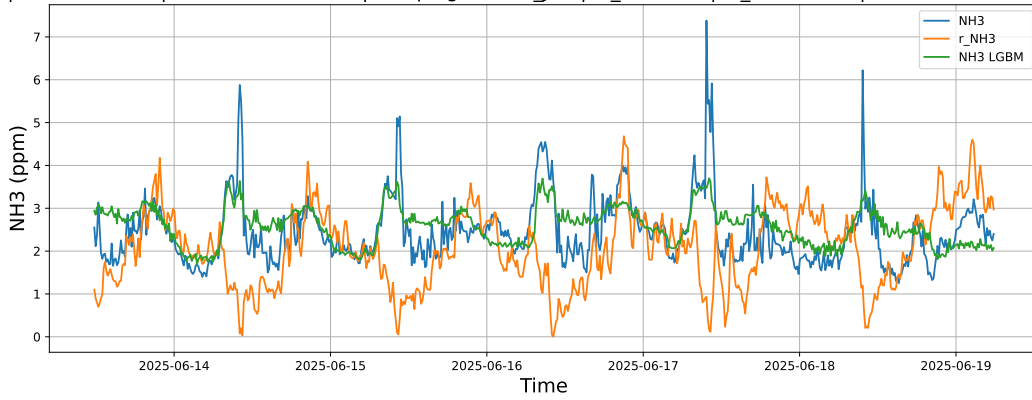


4.1.4.3 Time window : ALL

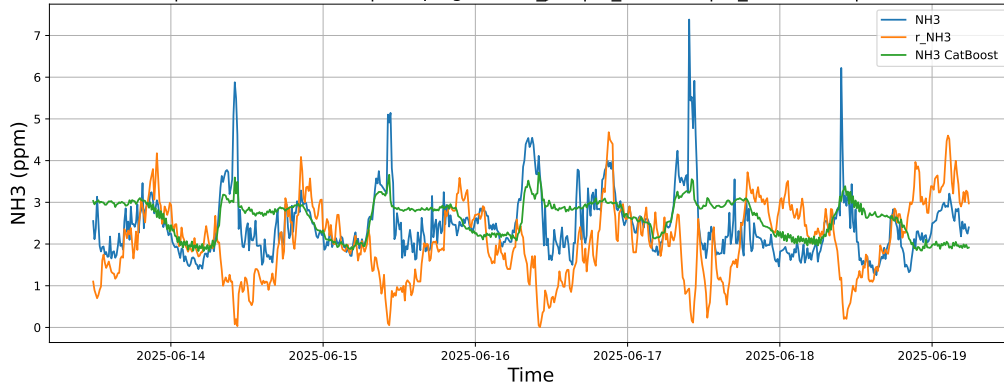
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



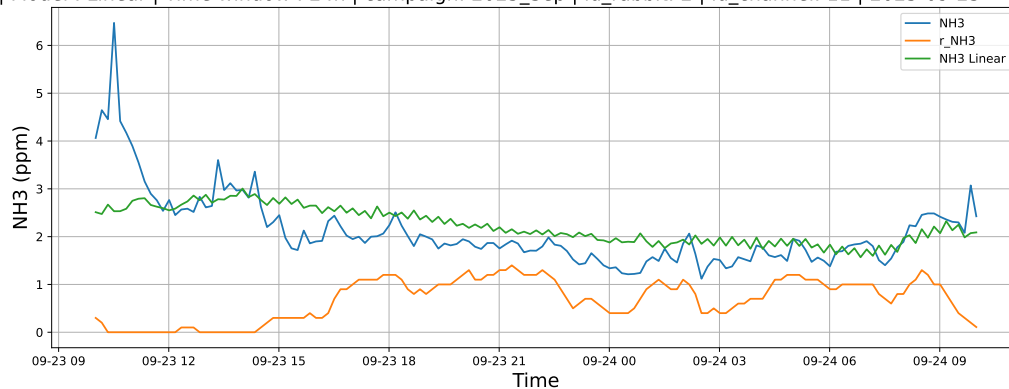
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



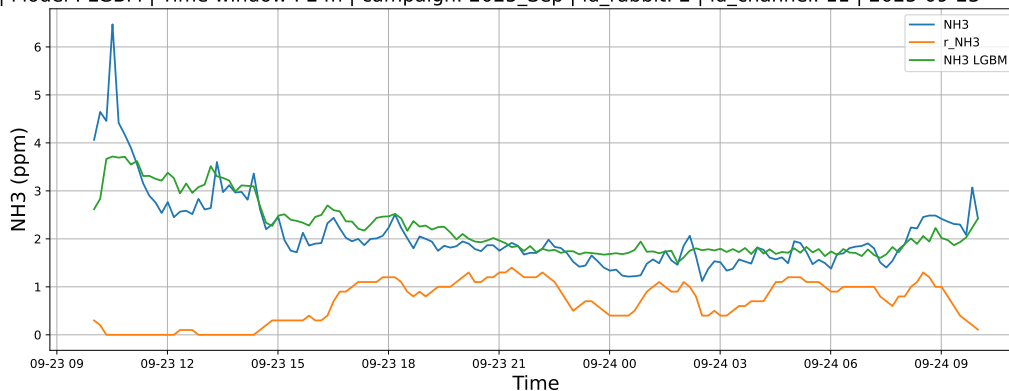
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

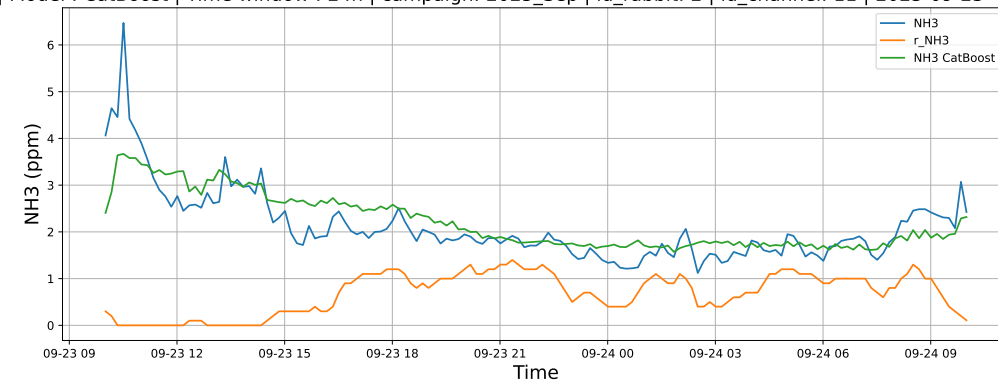
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

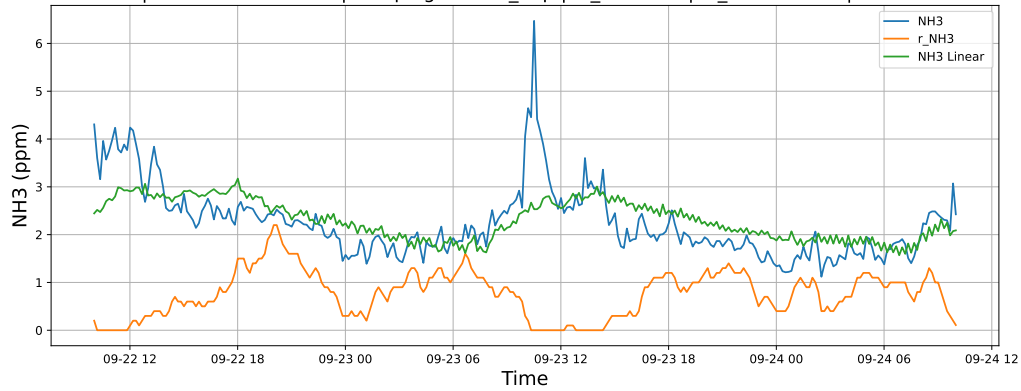


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

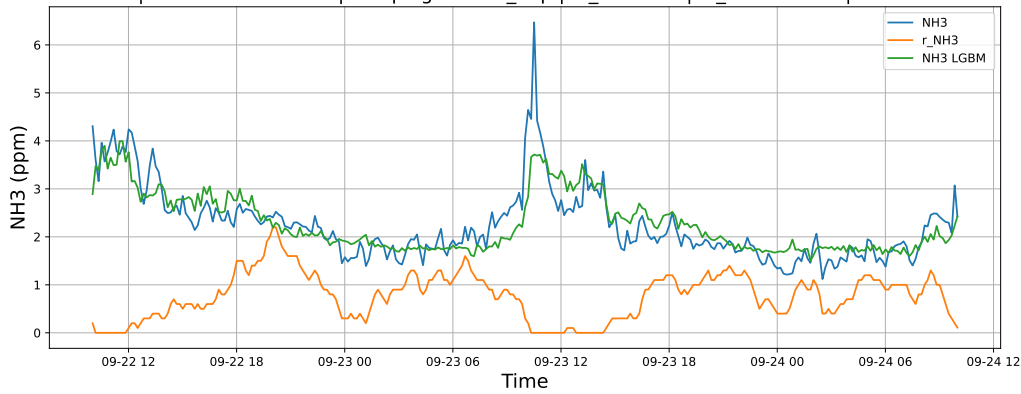


4.1.5.2 Time window : 48

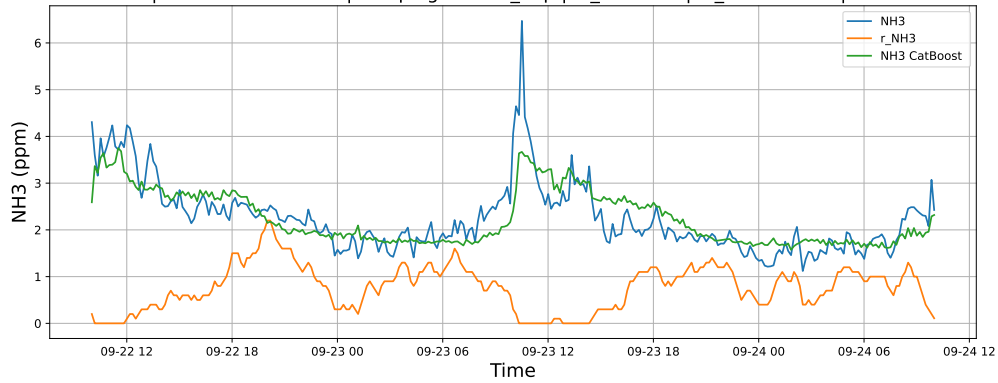
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

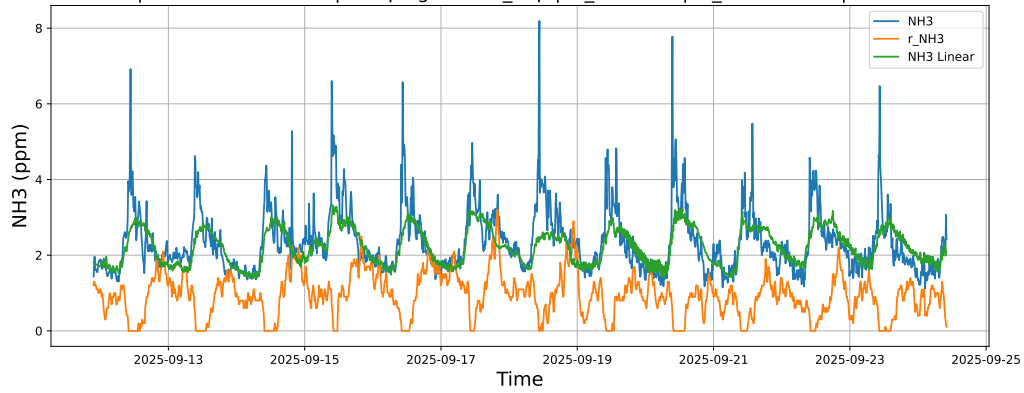


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

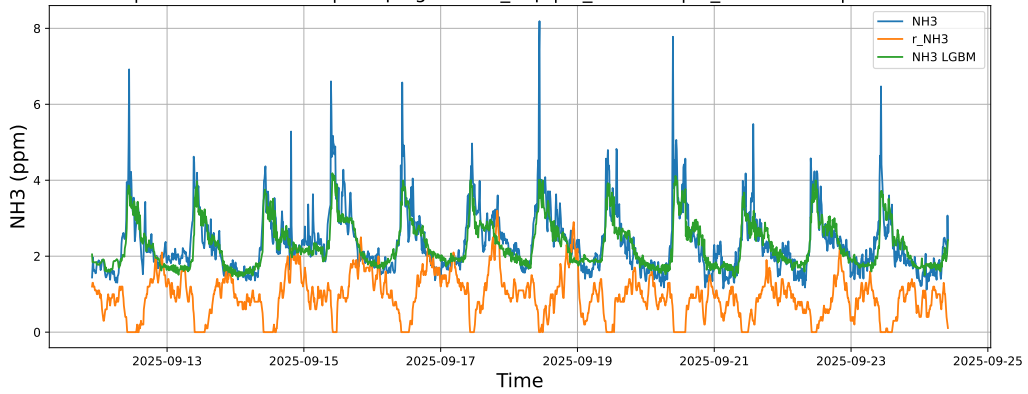


4.1.5.3 Time window : ALL

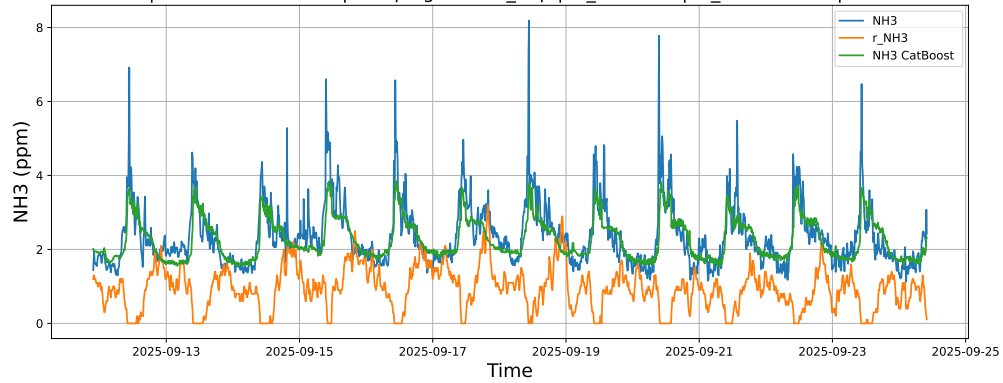
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



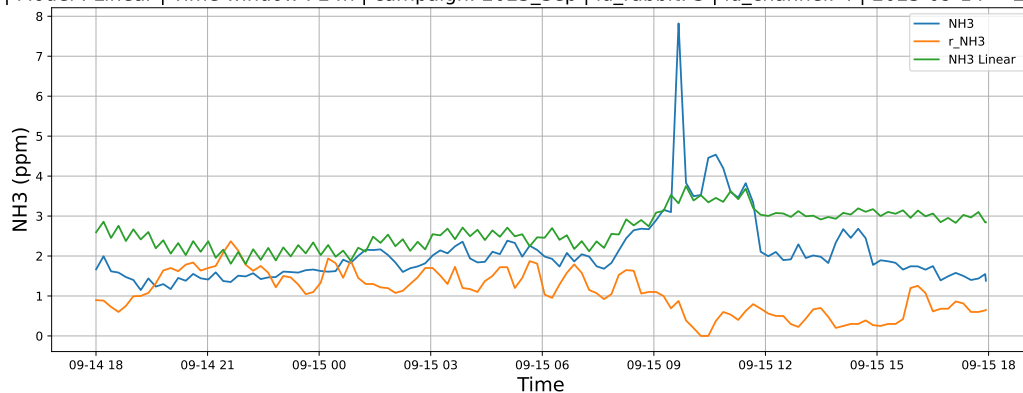
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



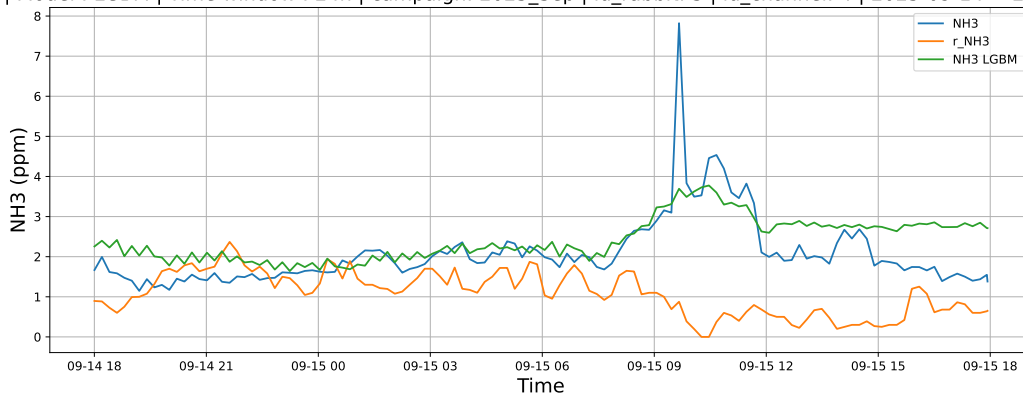
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

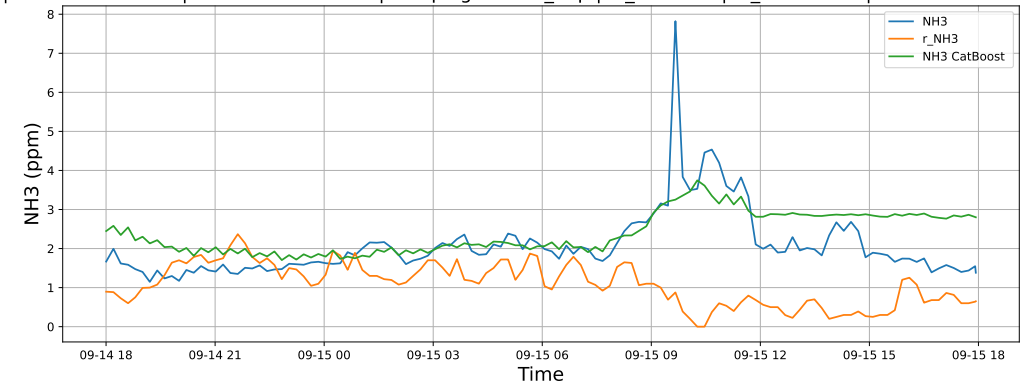
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

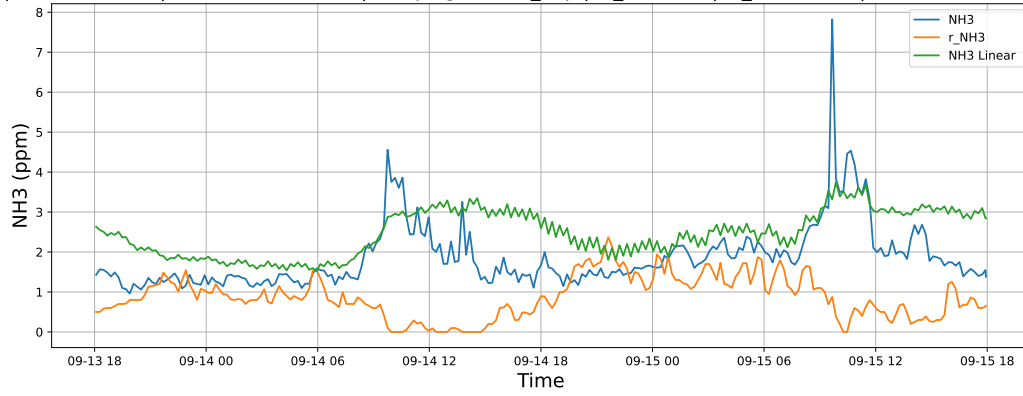


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

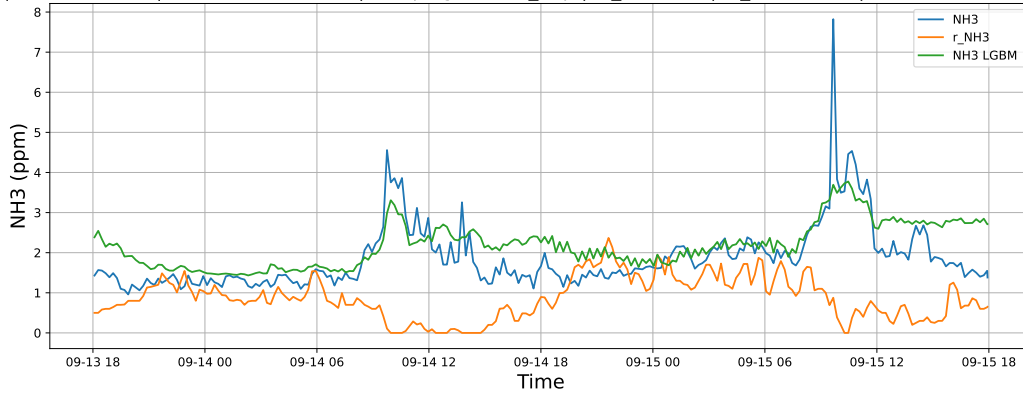


4.1.6.2 Time window : 48

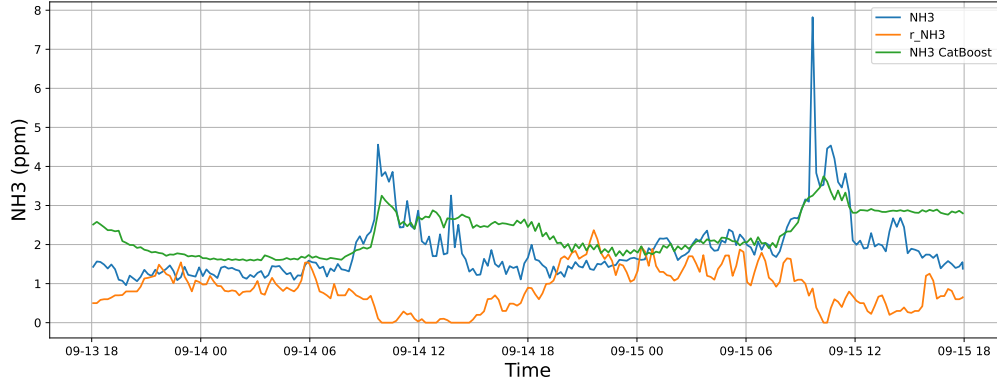
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

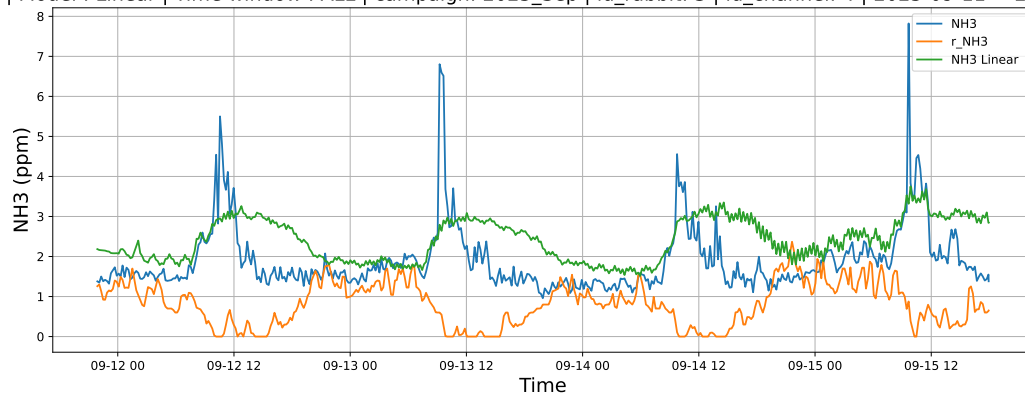


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

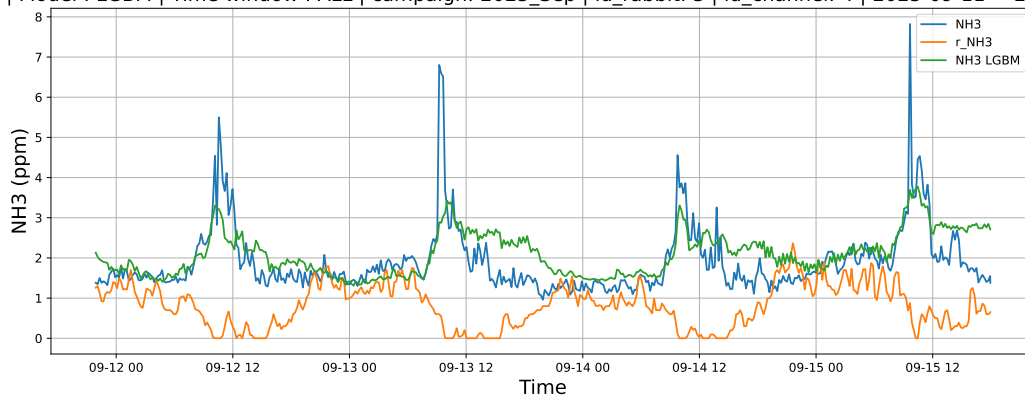


4.1.6.3 Time window : ALL

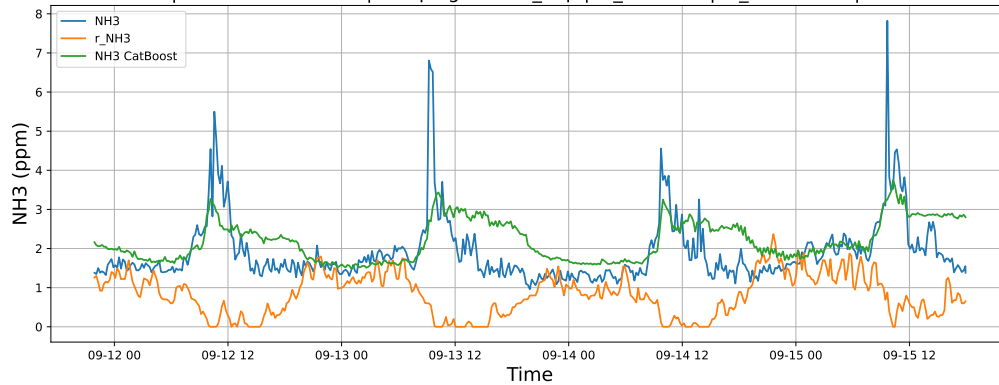
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



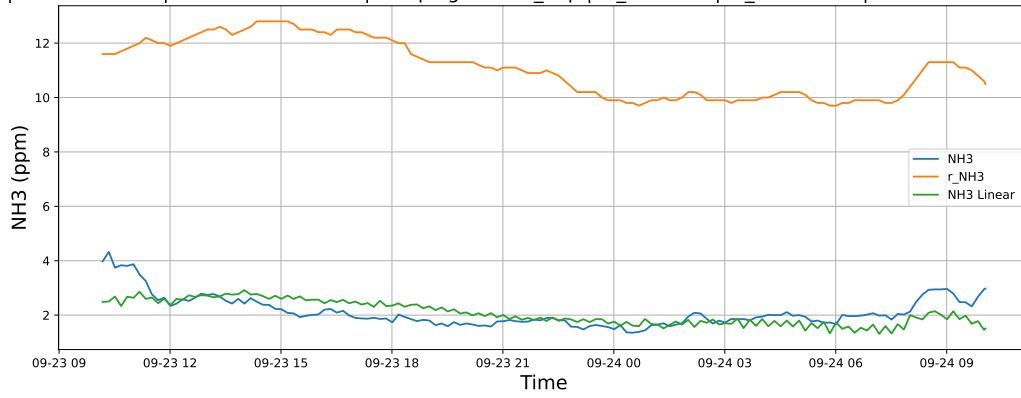
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



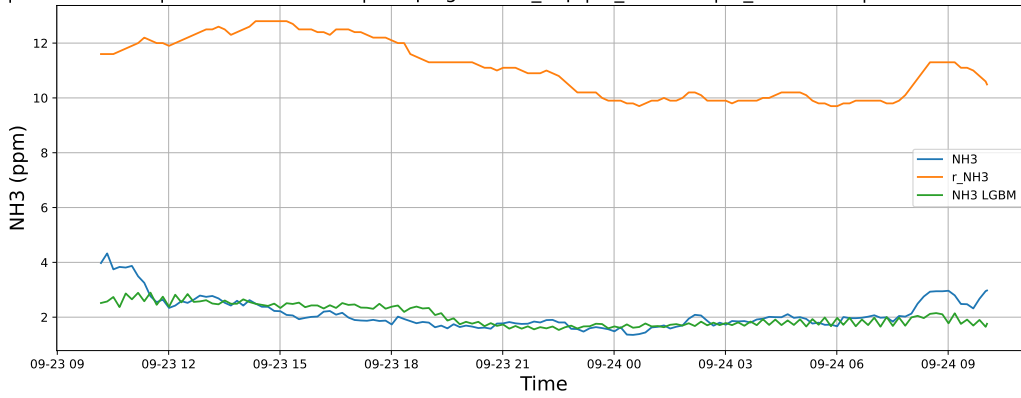
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

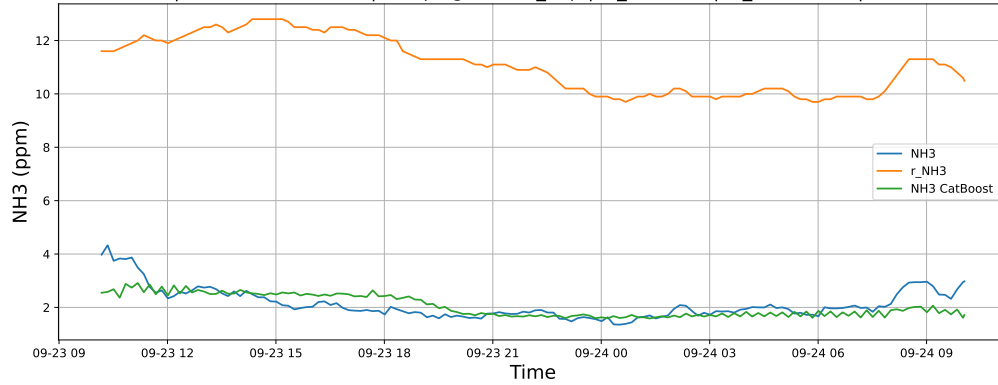
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

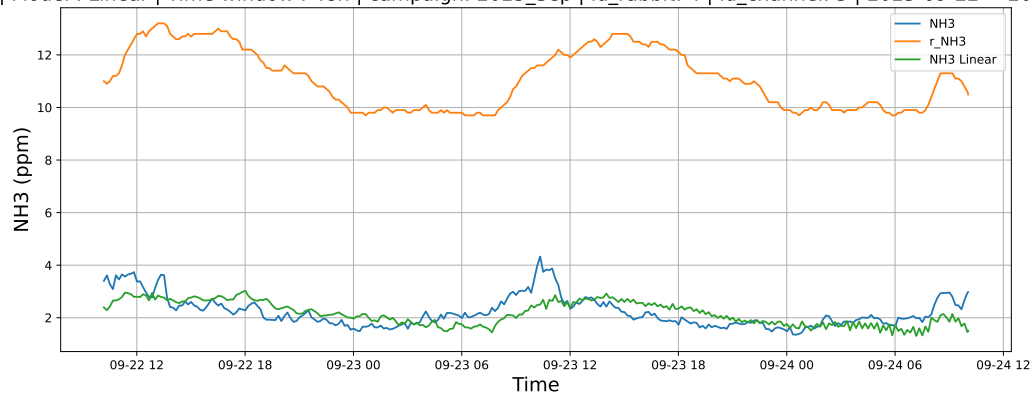


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

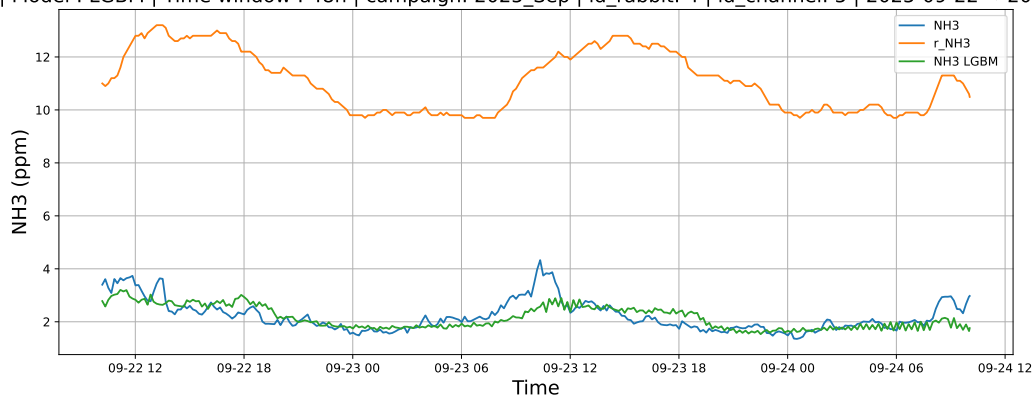


4.1.7.2 Time window : 48

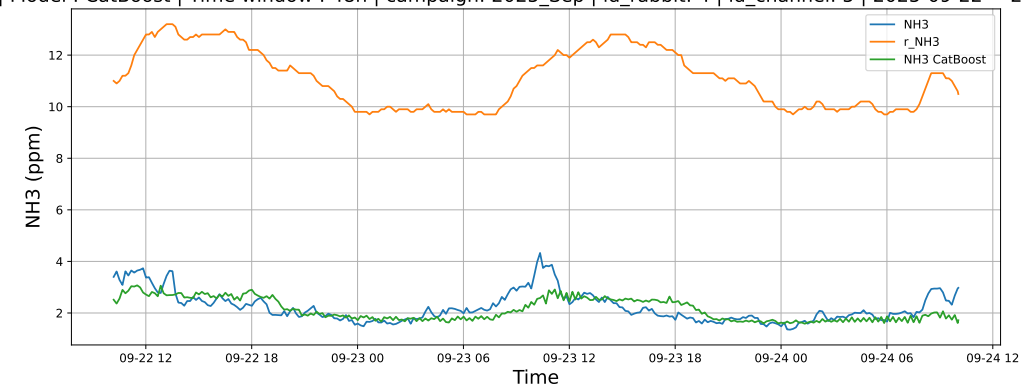
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

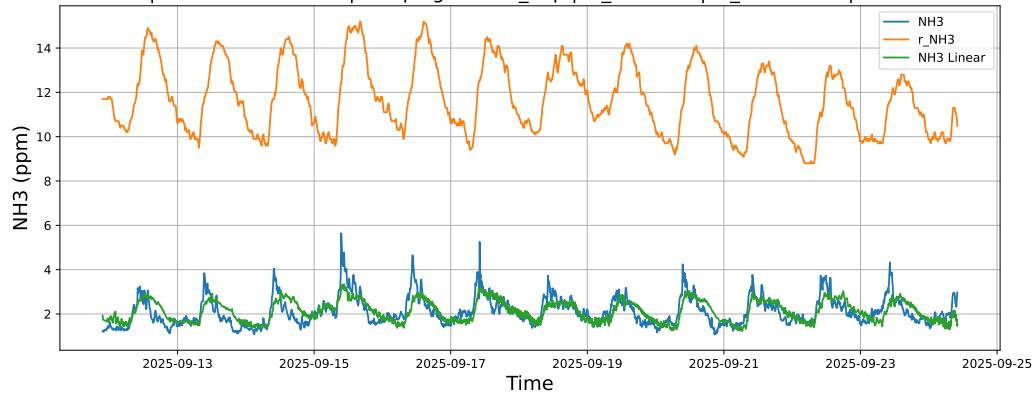


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

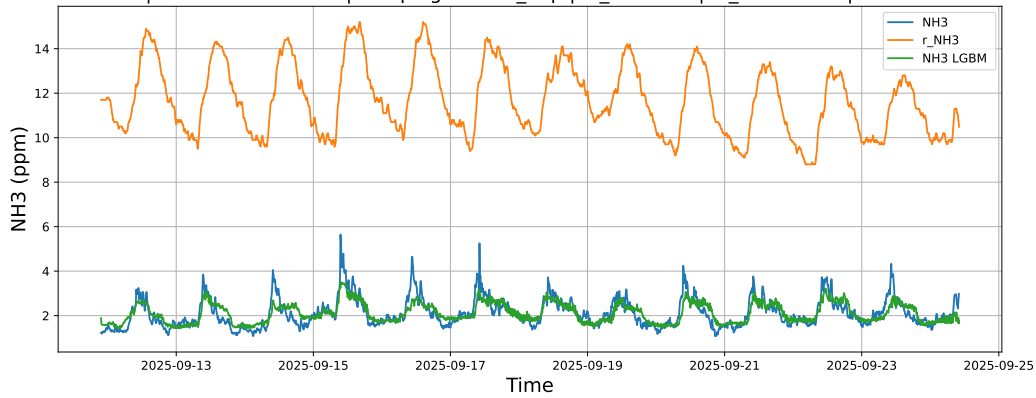


4.1.7.3 Time window : ALL

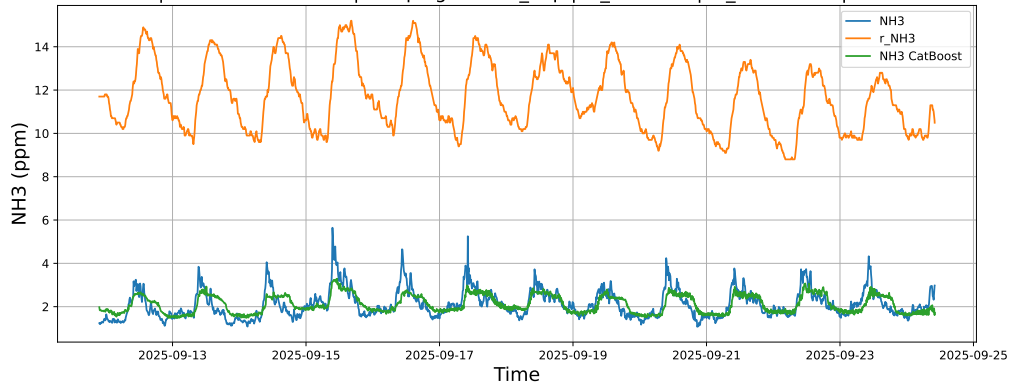
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



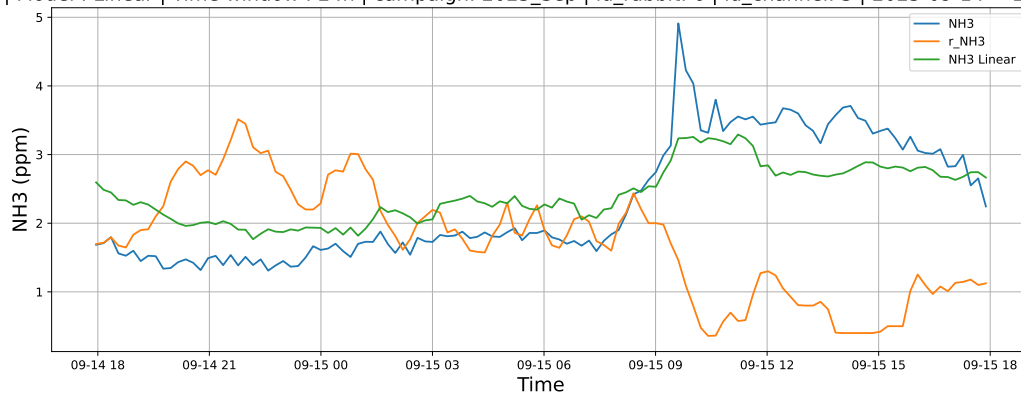
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



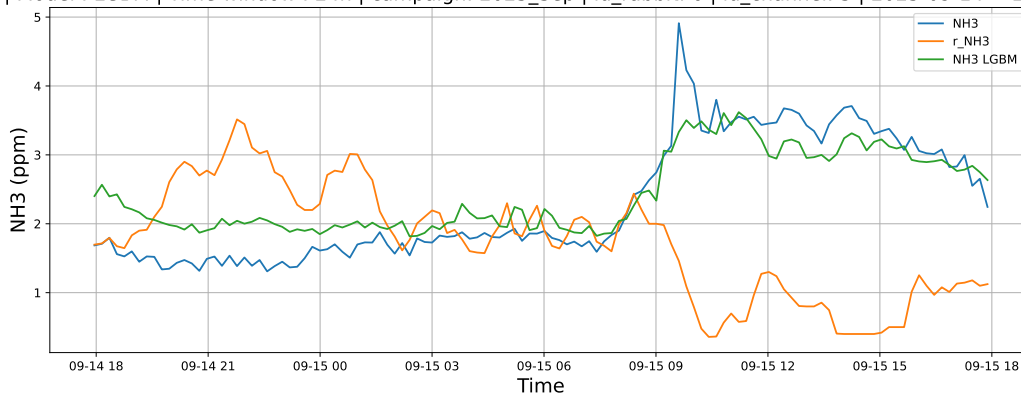
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

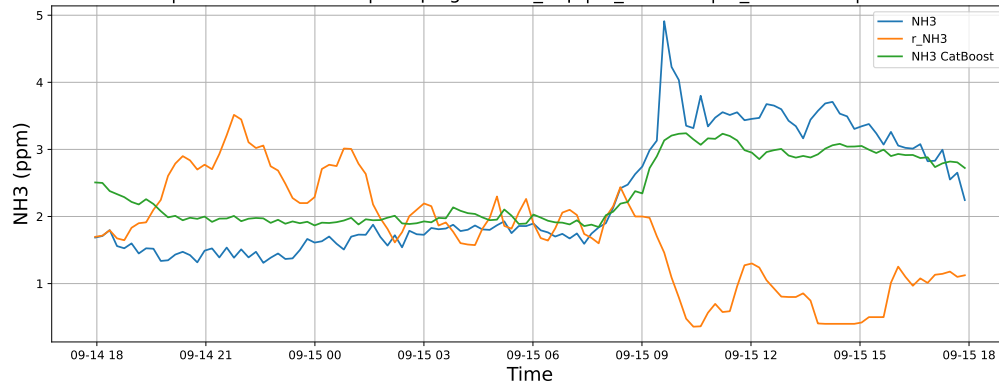
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

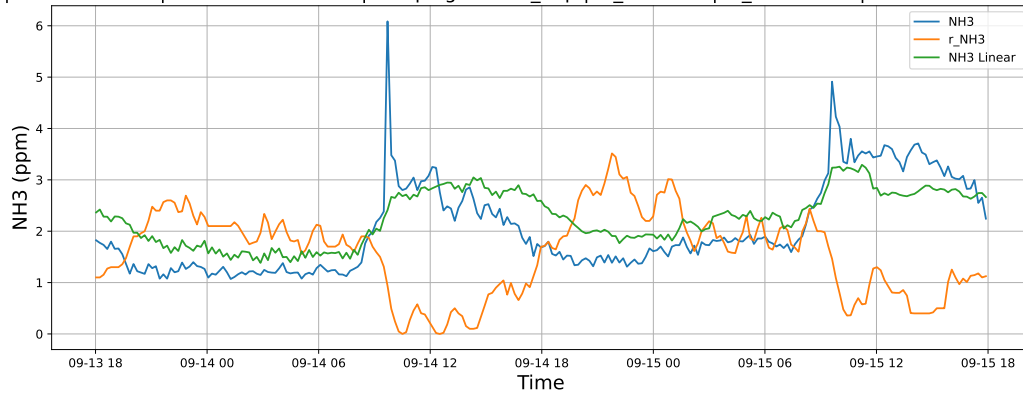


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

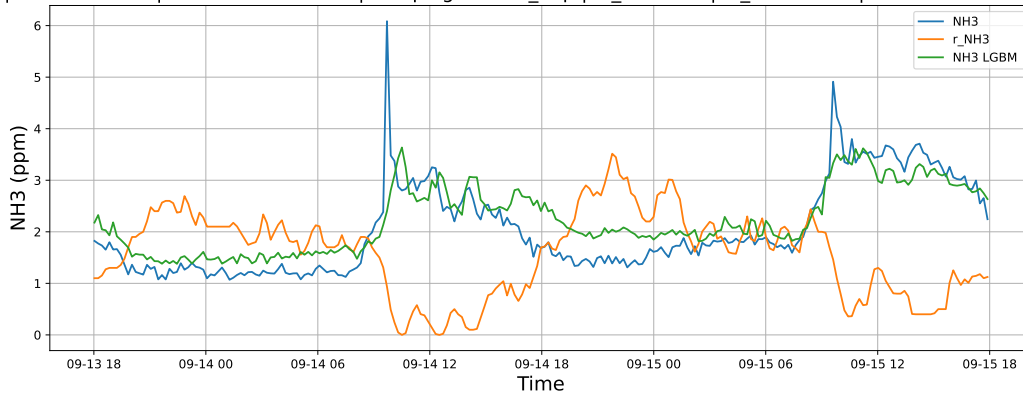


4.1.8.2 Time window : 48

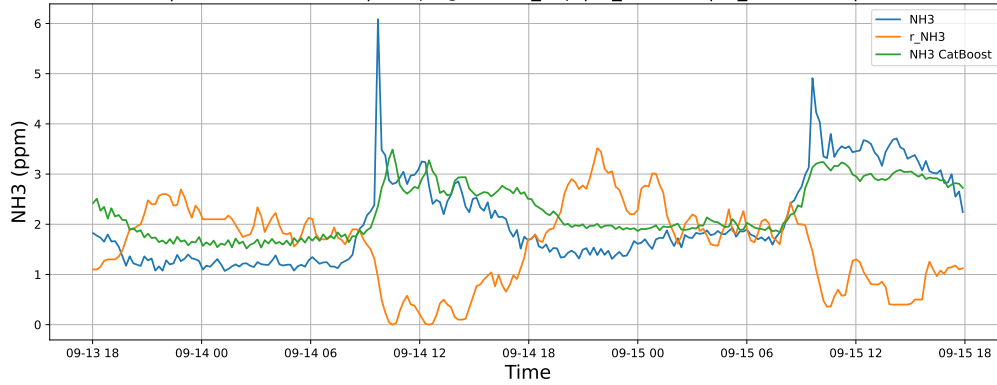
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

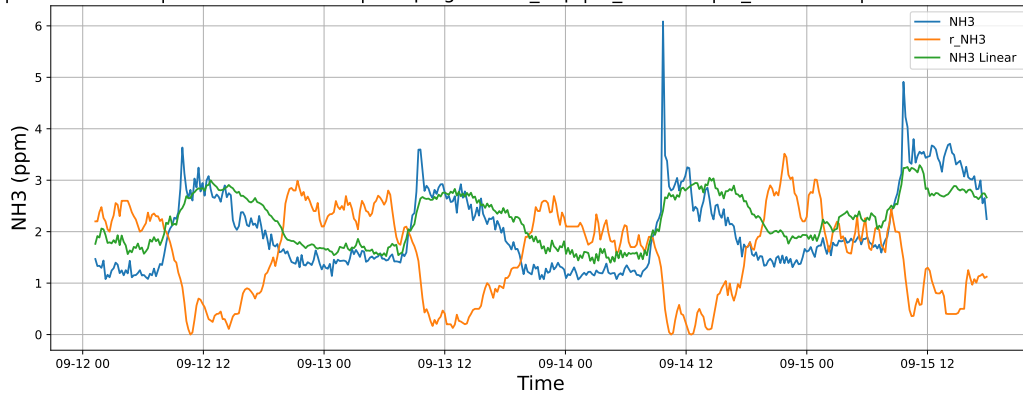


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

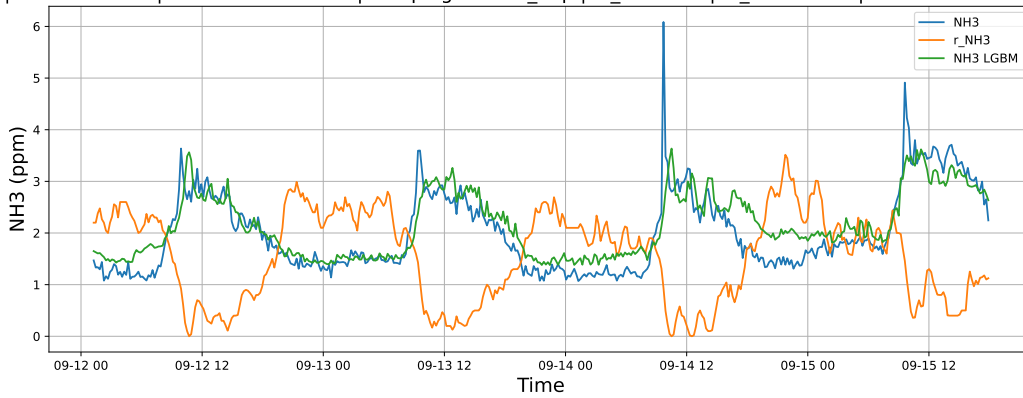


4.1.8.3 Time window : ALL

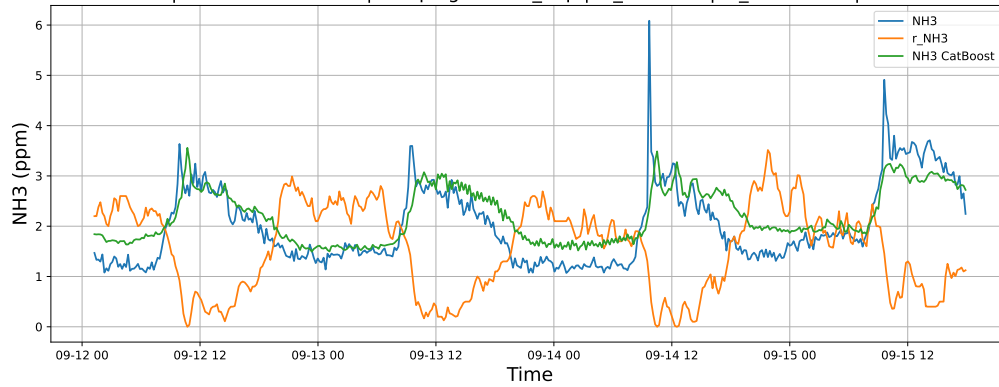
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-12 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-12 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-12 → 2025-09-15

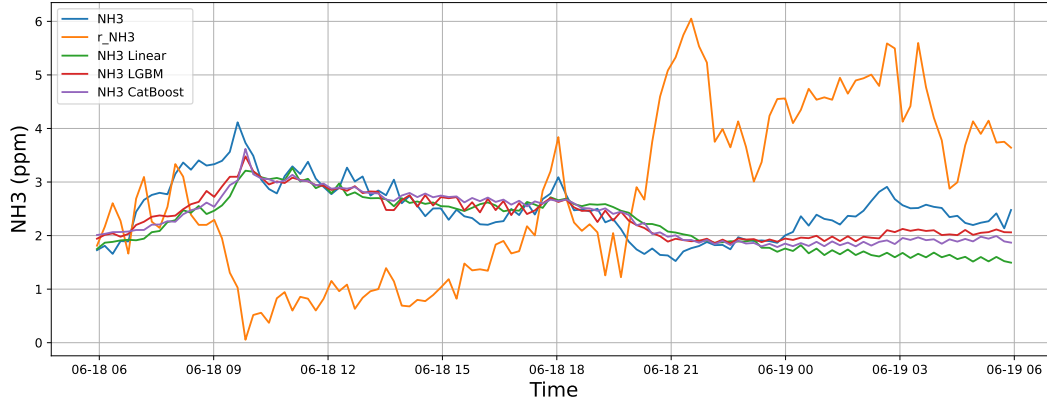


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

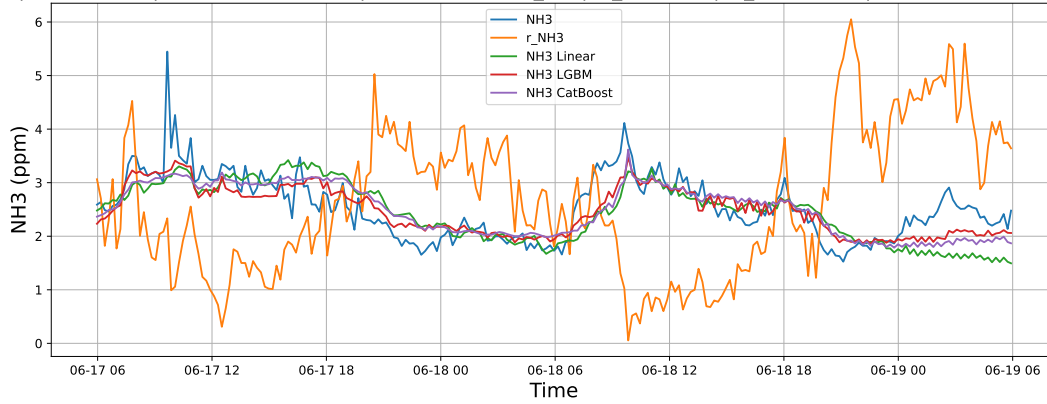
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



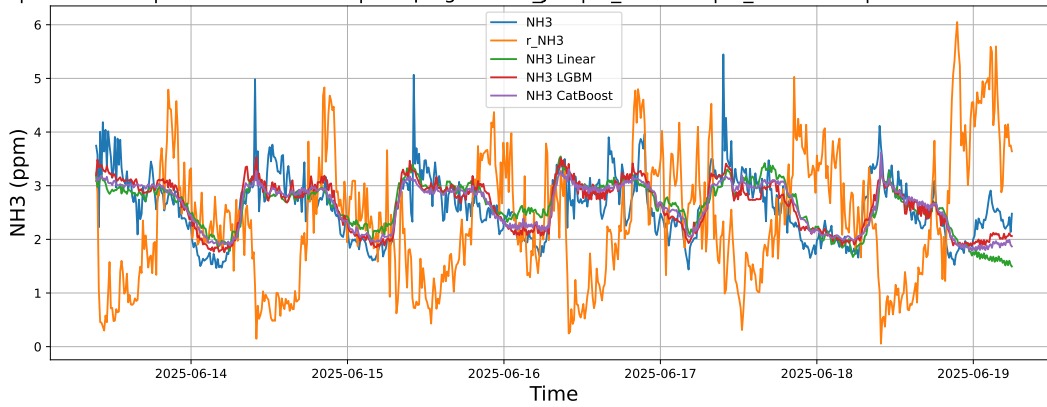
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

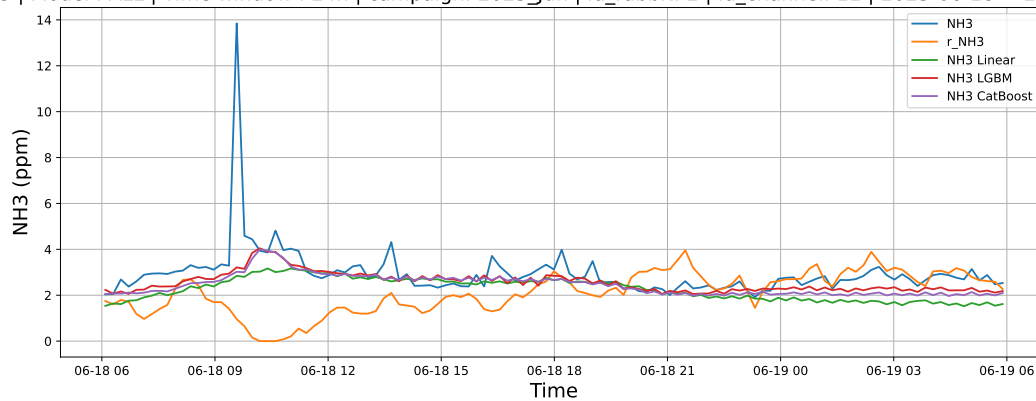
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

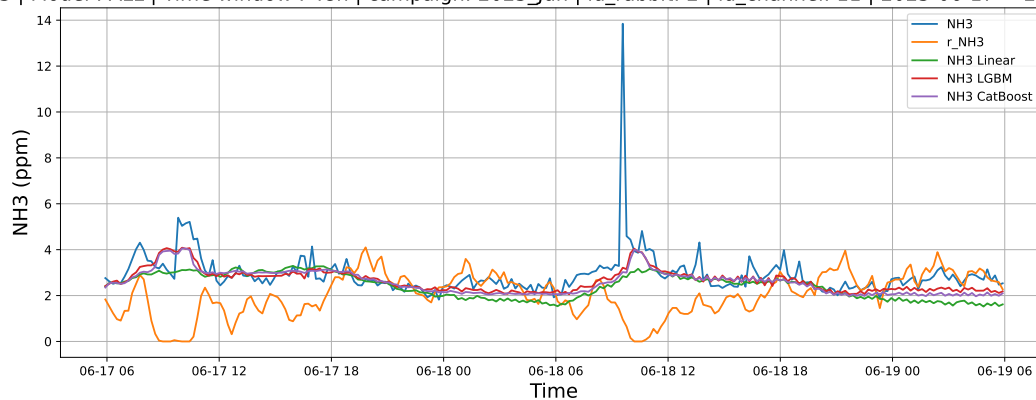
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



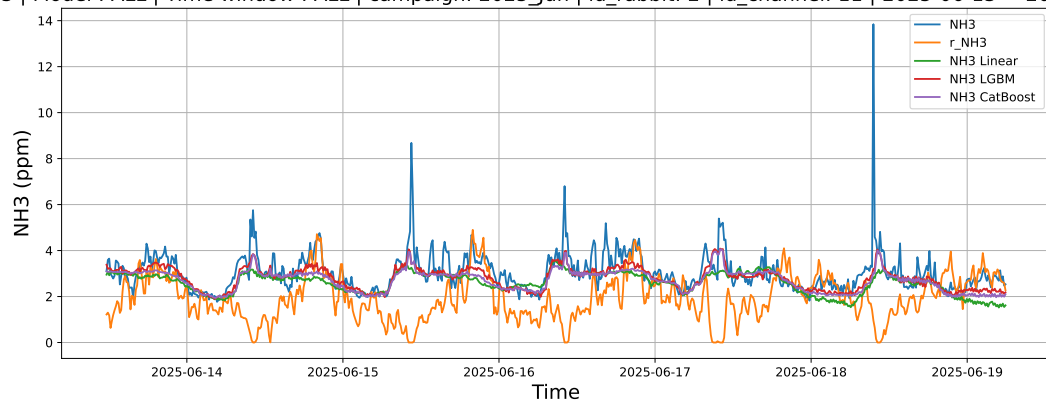
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

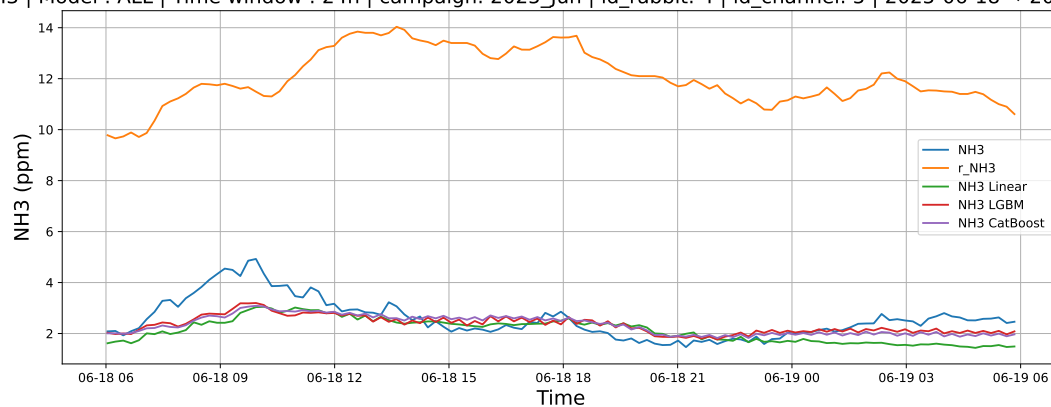
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

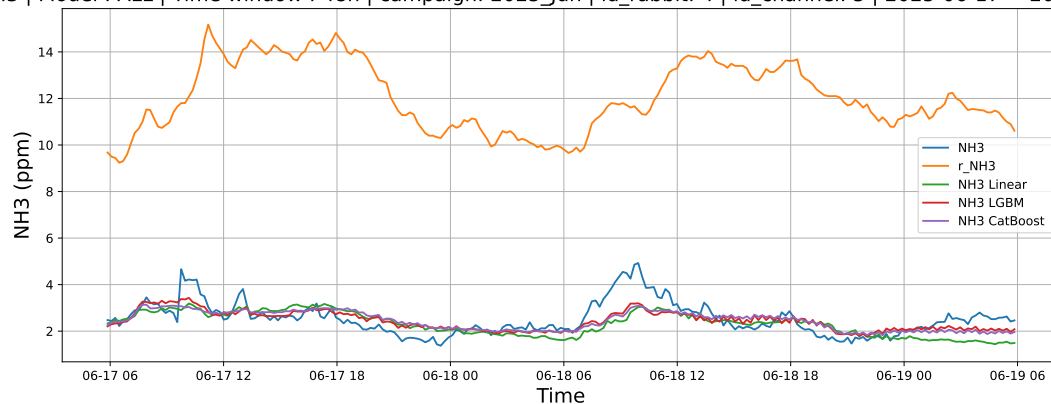
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



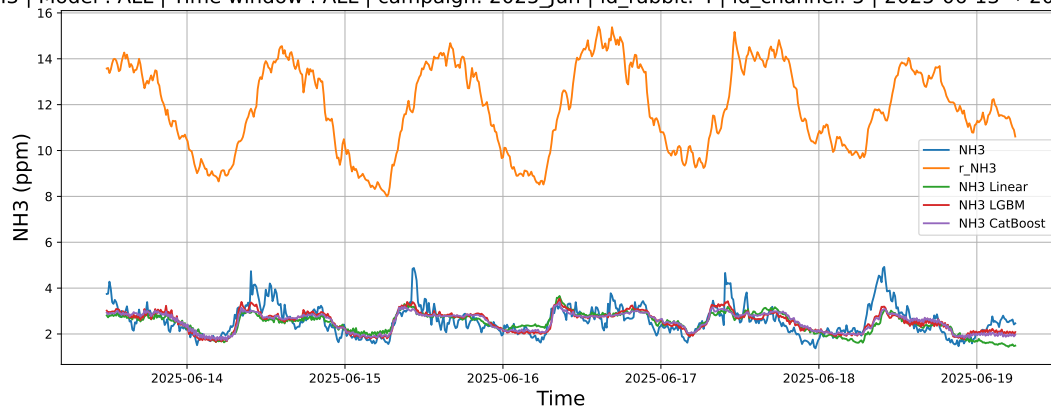
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

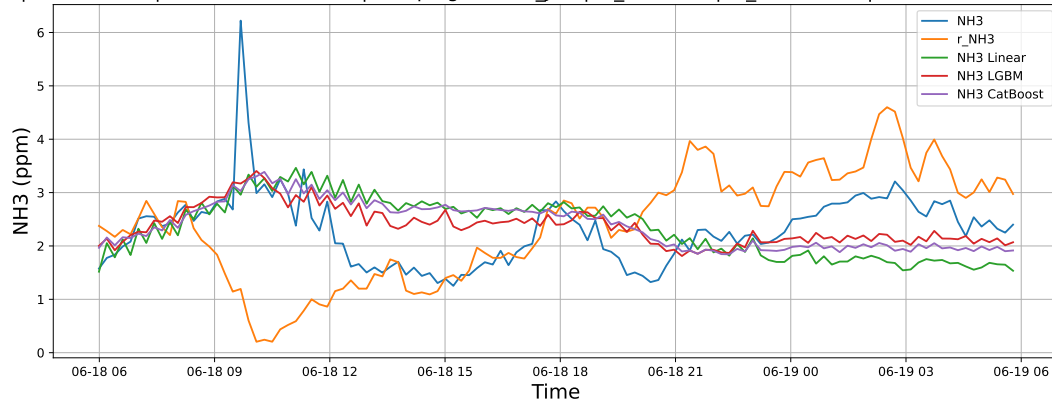
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

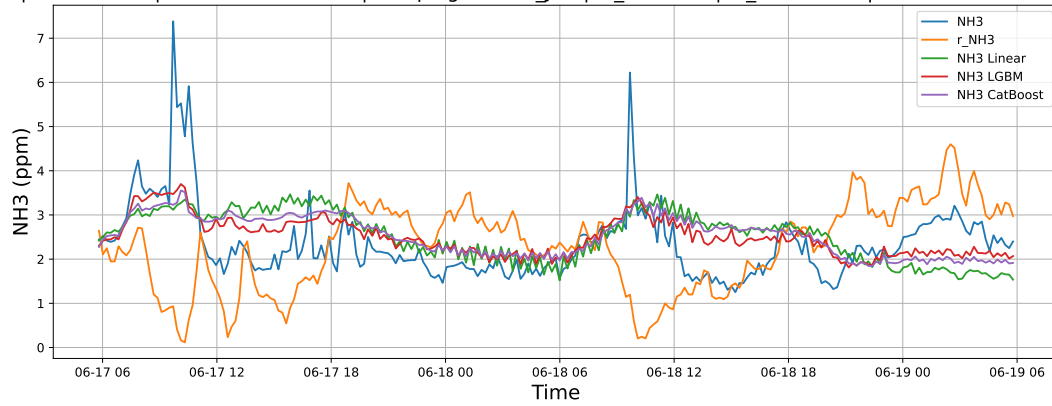
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



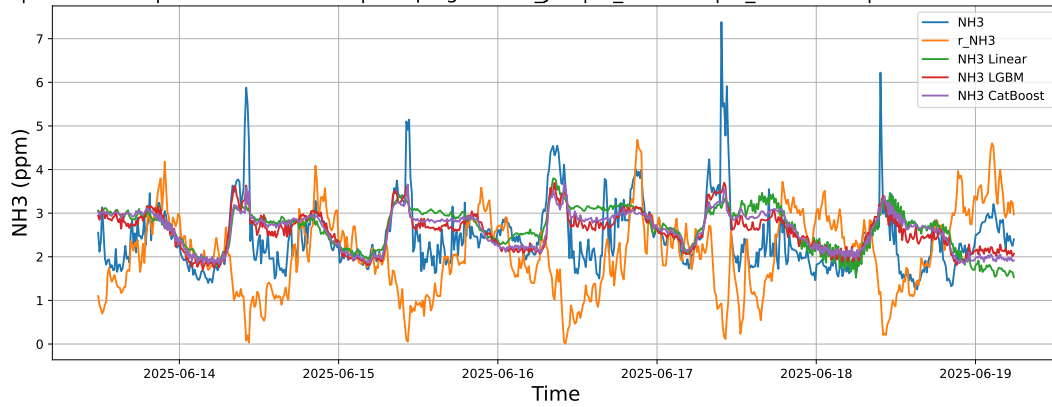
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

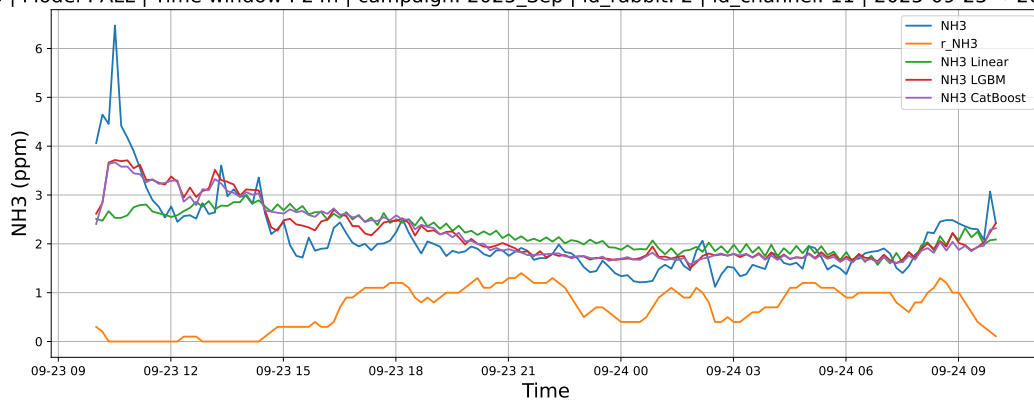
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

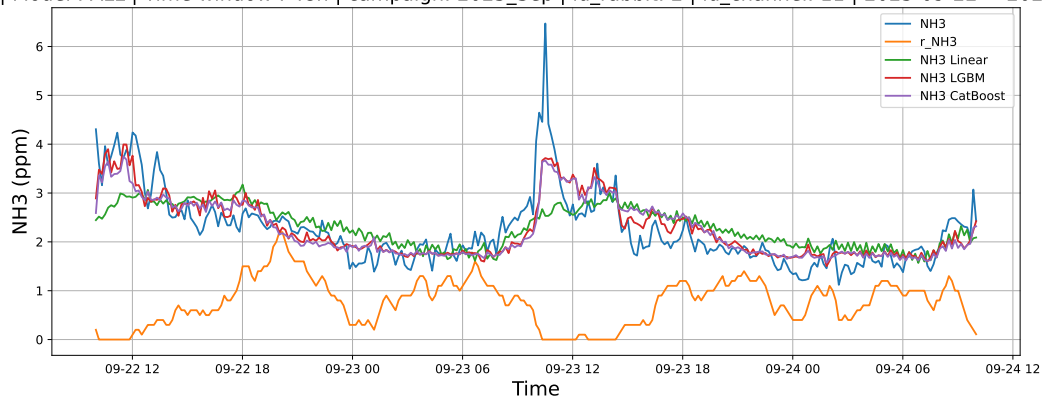
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



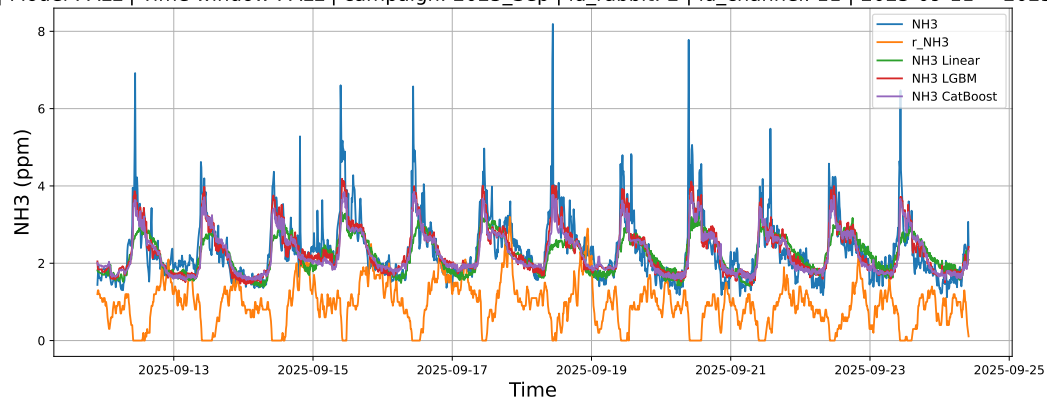
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

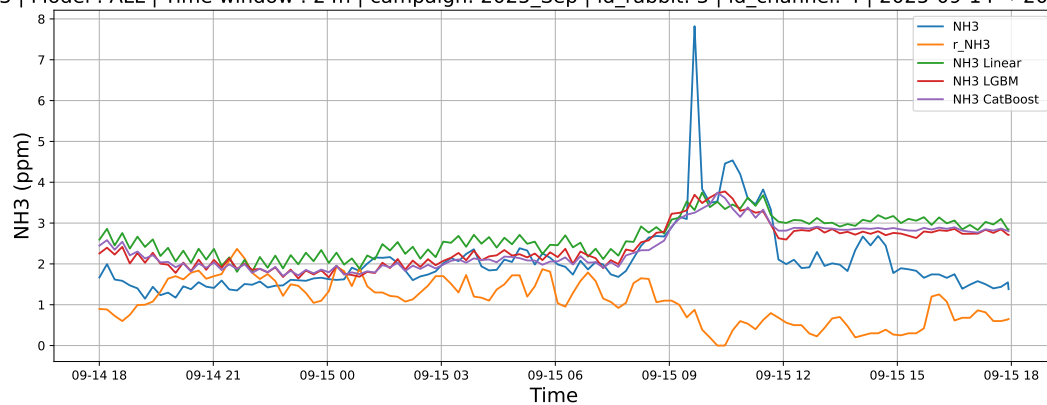
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

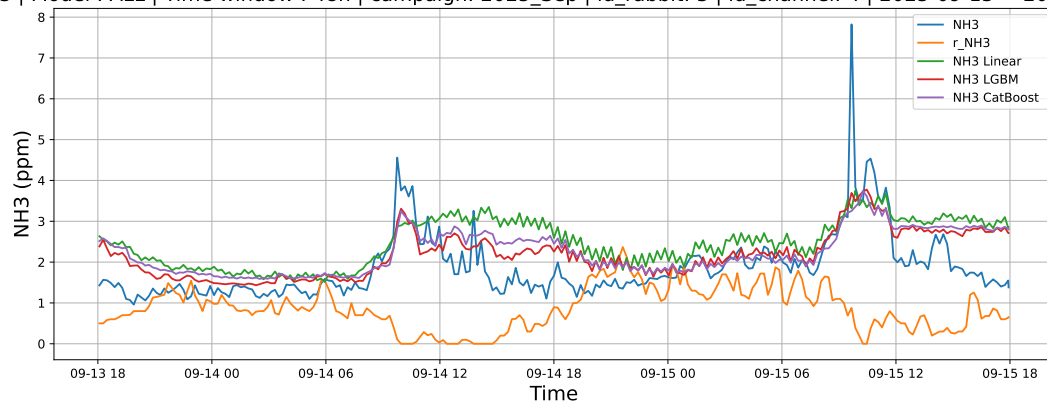
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



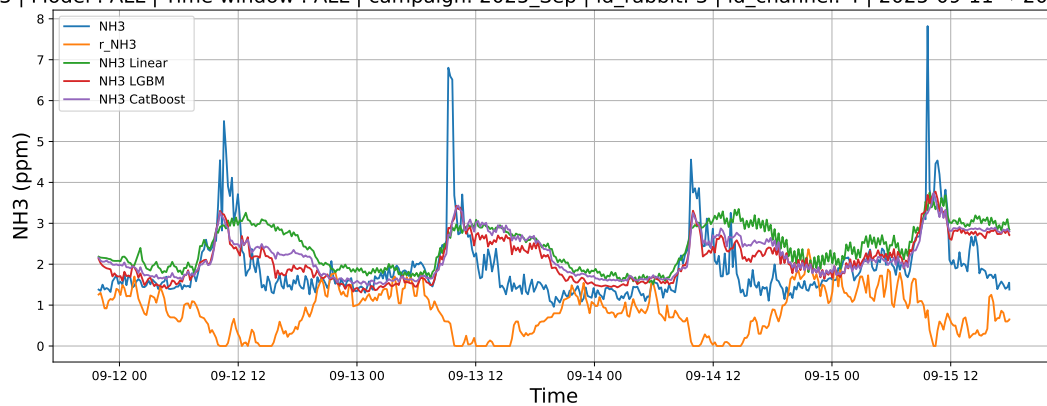
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

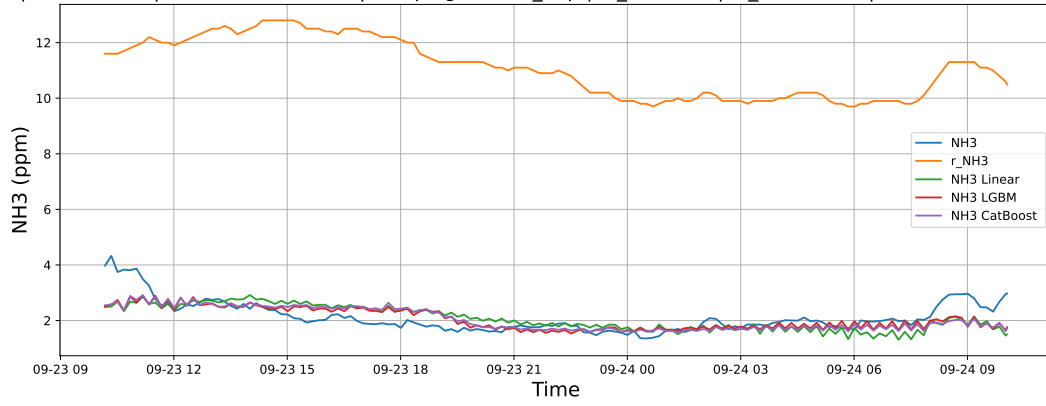
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

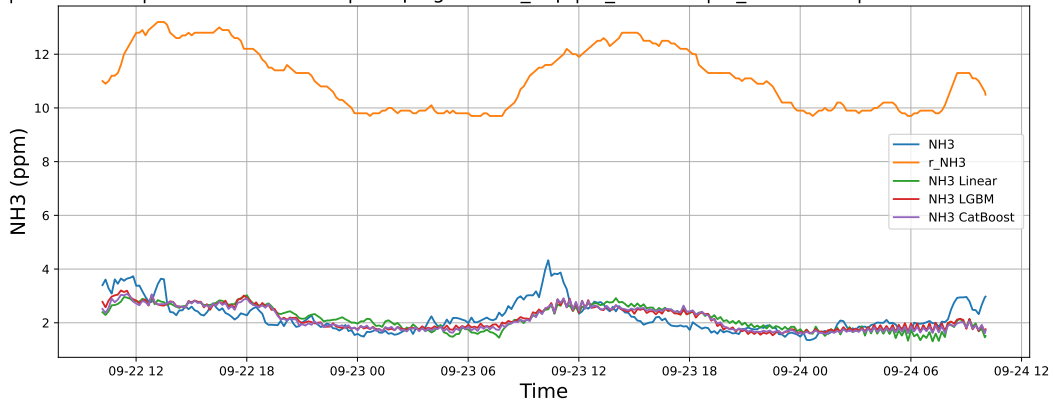
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



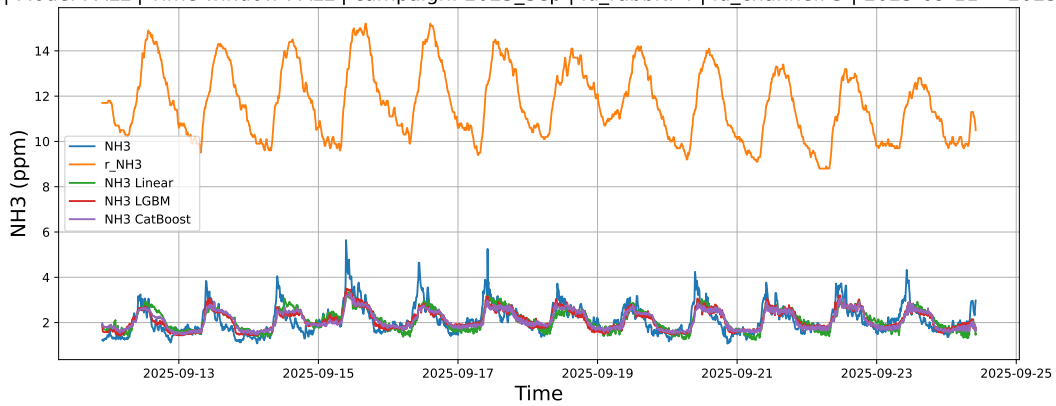
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

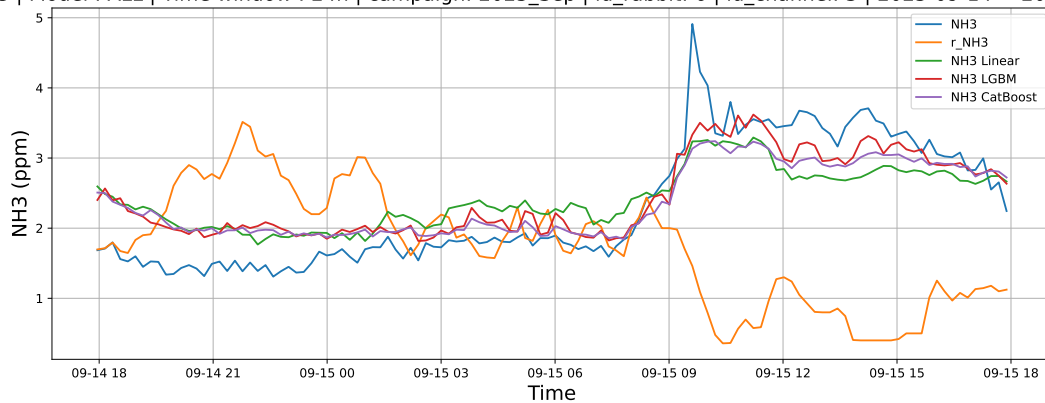
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

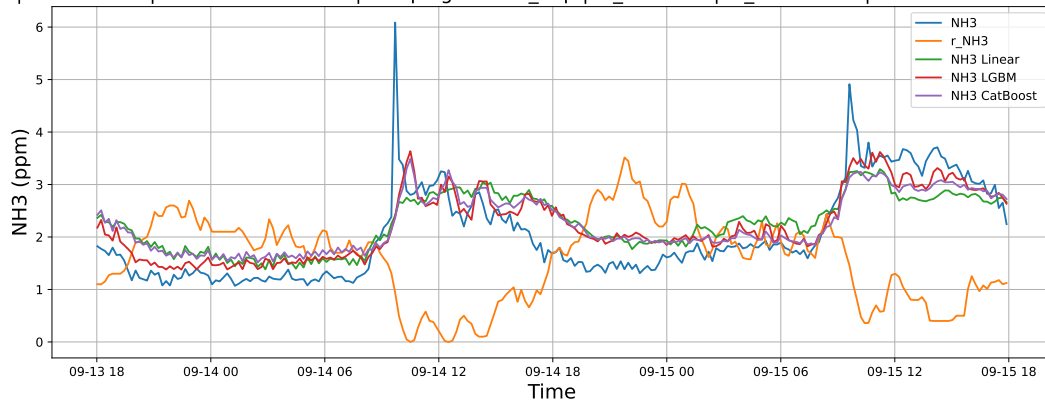
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-12 → 2025-09-15

